

Precision Flow Conditioner for Reproducible Cavitation & LENR-Related Experiments

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Open Hardware Project — 2026
Inspired by the work of Bob Greenyer and the
Martin Fleischmann Memorial Project (MFMP)

Complete Sphinx Documentation • Single Volume Edition

CRITICAL DISCLAIMER

*Performance graphs, bubble-size estimates, and all quantitative models in this document are **conceptual software estimates only**. The author has not built or tested a physical prototype. Nothing here constitutes validated engineering data. Use at your own risk.*

⚠ Important Notice

Before using this project, please read the [Project Status & Limitations](#) page.

Performance graphs and models are conceptual estimates only. There is currently no experimental validation by the author.

A fully parametric, 3D-printable, open-hardware device for repeatable hydrodynamic cavitation experiments.

Built around a stack of 25 highly polished Enterprise 3.5” HDD platters (Tesla-style boundary layer spiral) with an optional central Hybrid VAWT (HVAT) flow straightener. Designed specifically as a precision gas/liquid injection tool for LENR and cavitation research in the tradition of Bob Greenyer and the Martin Fleischmann Memorial Project (MFMP).

Project Status & Limitations

READ THIS PAGE BEFORE USING, BUILDING, OR CITING ANYTHING FROM THIS PROJECT.

This is not a finished scientific instrument.

This is not a validated testbed.

The performance models are not predictions.

If you are looking for proven hardware with reliable data, **this is not it yet.**

If you proceed anyway, you do so with full knowledge that almost everything performance-related here is speculative.

Current Maturity Level

Maturity Matrix (as of May 2026):

Area	Maturity Level	Can You Trust It?	Notes
Physical CAD & BOM	Good	Yes (for building)	Designs are buildable on paper
3D Printing Documentation	Good	Mostly	Practical guidance exists
Assembly Instructions	Good	Mostly	Jigs and procedures are detailed
Engineering Drawings & Visuals	High	For visualization	Pretty pictures, many are generated
Performance Graphs & Models	Very Low	No	Simplified speculation only
Experimental Validation	None	No	Author has not built or tested anything
Scientific Usefulness as Testbed	Unknown	No	Unproven. This is the entire open question
Reputation as “Consistent Testbed”	Does Not Exist	Do not claim	No data exists to support this yet

Bottom line: You can build something that looks like the drawings. Whether it actually delivers better, more consistent cavitation conditions than simpler methods is currently unknown.

Critical Disclaimers (Read These Carefully)

1. Performance Graphs and Models — These Are Speculation

All performance curves, “what-if” scenarios, bubble size estimates, swirl reduction plots, hybrid synergy diagrams, pressure drop estimates, etc. are:

Crude, back-of-the-envelope analytical guesses.

Not CFD.

Not based on any physical experiments by the author.

Not validated against real data.

They were created to explore ideas and generate hypotheses. They should **not** be treated as reliable indicators of what will actually happen when you build the device.

Do not cite these graphs in papers. Do not present them as evidence. Doing so would be misleading.

The author makes no claim that these models accurately predict real-world behavior. They are educational toys, not scientific tools.

2. No Experimental Data From the Author

The author has not built or tested this device.

As of now, there are zero physical prototypes built by George Lambert, zero measurements, and zero data.

Everything written about how the device “will” or “is expected to” perform is either:

Theoretical reasoning, or

Output from very simple models that the author himself does not trust as accurate.

If you see anyone (including the author in the future) claiming specific performance numbers from this project without showing real build data, treat those claims with extreme skepticism.

3. This Is an Open Design Exploration

This project should be viewed as:

An open hardware starting point for precision flow conditioning in cavitation experiments.

A well-documented set of tools and parametric designs.

An invitation for the community (especially the MFMP and broader cavitation research community) to build, test, improve, and validate.

It is **not** a finished, validated scientific instrument.

What This Project Actually Provides

Complete, parametric, open-source CAD (OpenSCAD) for two variants (HDD platters and CD/DVD discs).

Extensive documentation on design rationale, assembly, and 3D printing.

Useful tooling for generating engineering drawings, DXF fabrication files, flow visualizations, and calibration aids.

A framework for systematic experimentation (calibration pieces, example configurations, etc.).

What This Project Does NOT Provide (And May Never Provide)

Any validated performance data

CFD validation of the designs

Experimental protocols that have been shown to work

Peer-reviewed findings

Independently replicated results

A “consistent testbed” that has actually been proven consistent

There is currently no evidence that this device works better than simpler injection methods.

Any such claim is currently speculation.

Recommendation for Users and Replicators (Blunt Version)

If you build this device:

Do **not** assume it will perform better than a simple orifice or bubbler. It might. It might not. Nobody knows yet.

Do **not** cite the graphs in this project as justification for your results.

Do **not** present this design as a “proven consistent testbed.” It is not proven.

Document everything honestly, including failures and null results.

If your results look exciting, be extremely cautious about causation vs correlation until you have rigorous controls and independent replication.

Publishing exciting results based on unvalidated hardware from this project would be premature and potentially damaging to the credibility of the field.

How to Cite This Project (Do It Honestly)

When referring to this work, use language similar to one of the following:

Recommended (accurate):

“The Precision Flow Conditioner is an open hardware design exploration by George Lambert (2026). The physical designs are available for replication, but as of publication, no experimental validation or performance data from the author exists. All performance estimates in the documentation are simplified conceptual models only.”

Avoid (misleading):

Claiming this provides a “consistent testbed”

Citing the performance graphs as evidence

Presenting the device as proven superior to simpler injection methods

Misrepresenting the current state of this project in papers or presentations would be scientifically irresponsible.

Project Philosophy

This project was created in the spirit of open, low-cost, high-documentation experimental work championed by Bob Greenyer and the MFMP community. The goal is to lower the barrier to high-quality, repeatable cavitation experiments — not to make premature claims.

Transparency about limitations is part of that philosophy.

Last updated: May 2026

If you have built this device or have feedback on these limitations, please open an issue or contact the author. Real-world data from builders will be the most valuable contribution to this project.

Quickstart: Im-Impatient-So-Just-Get-Me-Started

Minimum you need to know to actually build and run this thing.

⚠ Warning: All performance claims and graphs in this project are simplified conceptual models. The author has not yet built or tested a physical version. Treat everything as a starting point for your own experiments. See [Project Status & Limitations](#) 2 for full disclaimers.

What You Are Actually Building

A **precision flow conditioner**, not a power turbine.

It takes messy inlet flow (air or liquid) and turns it into a clean, low-turbulence, spiraling jet using stacked smooth discs (inspired by Tesla's boundary-layer turbine). The goal is to create a more repeatable and characterizable bubble cloud when this jet hits your cavitation chamber.

There are two versions:

HDD Platter version — better performance, harder to source

CD/DVD version — cheap and easy to source (recommended for first builds)

Full Bill of Materials (Default 25-Disc HDD Version)

Item	Qty	Material	Notes	Source
Enterprise 3.5" HDD platters (95 mm)	25	Polished aluminum	Match thickness, clean with IPA	Scrap/eBay Enterprise drives
PETG or ABS filament	~800 g	PETG preferred	0.2 mm layers, 30-40% infill	Your printer
M3/M4 stainless screws (30-60 mm)	12-16	Stainless	For clamping	Hardware store
M3/M4 nuts + washers	As needed	Stainless	-	Hardware store
Printed spacer rings (1.0 mm)	24	PETG	<code>SPACER_RING.scad</code>	3D print
Printed shipping inserts (1.4 mm)	26	PETG	<code>SHIPPING_INSERTS.scad</code>	3D print
Clear viewport plate (optional)	1	Clear PETG	<code>CLEAR_VIEWPORT.scad</code>	3D print
Solid top + bottom covers	2	PETG	<code>OPERATION_COVERS.scad</code>	3D print
Assembly jig	1	PETG/PLA	<code>ASSEMBLY_JIG.scad</code>	3D print
HVAT rotor (hybrid Savonius + Darrieus)	1	PETG	<code>HVAT_BLADES.scad</code>	3D print
HVAT brake disc + caliper	1 set	PETG	-	3D print
608ZZ bearings	2	Standard	8 mm ID	Skateboard/hardware
5-8 mm shaft (carbon fiber preferred)	~80 mm	Carbon fiber	For HVAT rotor	-
Printed bearing housers (upper + lower)	2	PETG	<code>BEARING_HOUSERS.scad</code>	3D print
Nozzle manifold	1	PETG	-	3D print
Cavitation jet nozzle (Venturi)	1	PETG	<code>CAVITATION_NOZZLE.scad</code>	3D print
Orifice inserts (6/8/10/12 mm)	4+	PETG	Start with 8 mm	3D print
Sensor mounts	4	PETG	For MPX5700 or similar	3D print
6 mm ID tubing	~1 m	Food-grade silicone/PVC	-	-
Hose barbs / push fittings	6-8	Brass/plastic	-	-
Threaded inserts (M3/M4)	As needed	Brass	-	-
Isopropyl alcohol 99%	500 ml+	-	For cleaning platters	-

Optional but recommended: Thrust washer, base/stand, longer diffuser extension.

Parts You Must 3D Print (Filenames)

HDD Platter Version (default):

`HDD_Platter_Cavitation_Conditioner.scad` → Main housing + nozzles + diffuser

HVAT_BLADES.scad → HVAT rotor

BEARING_HOUSERS.scad → Upper and lower bearing housers

SPACER_RING.scad → 24× 1.0 mm spacers

SHIPPING_INSERTS.scad → 26× 1.4 mm shipping inserts

ASSEMBLY_JIG.scad → Assembly jig (most important tool)

CLEAR_VIEWPORT.scad → Clear viewport (highly recommended)

OPERATION_COVERS.scad → Solid top + bottom jet covers

CAVITATION_NOZZLE.scad → Venturi nozzle + orifice inserts

CD/DVD Version (alternative, cheaper): All files are in `scad/cd_dvd_version/`

Assembly Diagrams

Recommended:

`illustrations/exploded_assembly.png` — Best overall exploded view

`illustrations/cutaways/full_assembly_cutaway_3d.png` — Shows internal flow path

`illustrations/cutaways/housing_cutaway_3d.png` — Shows platter stack + HVAT inside housing

These are also available as SVGs and in `docs/_static/`.

Minimum Assembly Instructions (HDD Version)

1. **Print the Assembly Jig** first.
2. **Clean all 25 platters** thoroughly with 99% isopropyl alcohol.
3. Build the disc stack **inside the jig**:
 - Bottom end plate
 - Alternate: platter → 1.0 mm spacer → platter (24 spacers)
 - Top end plate (or clear viewport for testing)
4. Clamp with long M3/M4 screws through the aligned holes. Tighten evenly.
5. Slide the clamped stack into the main housing.
6. Install HVAT rotor on shaft + bearings inside the central hub.
7. Mount bearing housers and brake caliper.
8. Attach nozzle manifold and cavitation nozzle (start with 8 mm orifice).
9. Mount the whole device vertically above or beside your cavitation chamber.

Start testing with low-pressure air through the clear viewport before introducing liquid.

Full detailed steps: see the [Build.....32](#) page in this documentation.

Print the calibration pieces before anything else.

Spacer thickness and bearing bore tolerances are make-or-break for this device.

Good luck. Build it. Test it. Tell us what actually happens.

Next steps after the quickstart:

Read the full [Design](#)..... 9 and [Theory](#).12 sections for the “why”.
Use the [Customization Guide](#) 21 to tune parameters.
Follow the detailed [Build](#)..... 32 and [Printing Guide](#).28 when you are ready to fabricate.
Review the complete [Bill of Materials](#)..... 26 and [Engineering Drawings](#).42.
The standalone version of this quickstart is also available as [Im-Impatient-So-Just-Get-Me-Started.md](#) ↗ in the repository root.

Design Rationale & Key Specifications

This page summarizes the engineering decisions captured during the May 2026 design conversation.

Goal Clarification

The device is **not** intended as a power-producing turbine. It is a **precision flow conditioner** whose sole job is to deliver exceptionally uniform, low-turbulence, spiraled gas or liquid from tangential inlets to a central exit that then feeds a cavitation chamber.

The original hybrid VAWT idea was retained in a modified form: a lightly loaded central **HVAT (Hybrid Vertical Axis Turbine)** rotor that uses residual swirl energy to further straighten and organize the exit flow rather than to extract net electrical power.

Why HDD Platters?

- Enterprise 3.5” platters (95 mm OD) from dead server drives are:
- Mirror-polished to sub-nanometer roughness — perfect boundary layer behavior
- Extremely consistent in thickness and flatness
- Free / very cheap in bulk
- Already have convenient central hub geometry and mounting holes

Geometry That Matters

Parameter	Recommended Value	Rationale
Nozzle angle (from tangent)	8–12° (10° default)	Strong inward spiral without excessive radial velocity or shock
Number of nozzles	6	Excellent azimuthal uniformity; easy 3D printing
Inter-platter gap	0.8–1.5 mm (1.0 mm default)	Good channel count + practical printing + strong multi-channel effect
HVAT rotor diameter	~28 mm	Fits the natural central opening; provides active straightening
Diffuser divergence	6–8° half-angle	Avoids flow separation while recovering pressure

Materials for Reproducibility

- All structural parts are designed to be printed in PETG or ABS on a consumer FDM machine. The only non-printed “exotic” components are:
- 25 matched Enterprise HDD platters (cleaned with IPA)
 - Two 608ZZ bearings (or ceramic)
 - M3/M4 stainless hardware
 - 5–8 mm shaft stock for the HVAT
- This keeps the entire device accessible to the open LENR community.

Relation to Greenyer / MFMP Research

The design directly attacks the “injection variability” problem in hydrodynamic and hybrid ultrasonic cavitation experiments. By providing a repeatable, multi-channel, low-turbulence spiral inlet + tunable HVAT + precision Venturi nozzle, experimenters can vary one parameter at a time (gas type, flow rate, residual swirl, ultrasonic power, foil material, deuterium concentration, etc.) while keeping the fluid dynamic boundary conditions far more constant than with a simple bubbler or horn.

This is exactly the spirit of the MFMP: radical accessibility + obsessive documentation + public data.

Future Directions (Open)

Servo-controlled variable-pitch HVAT

Integrated data logging (pressure + temperature + neutron/gamma)

Full OpenFOAM validation of the 10° / 6-nozzle baseline

Larger (40+ platter) high-throughput version

Metal (aluminum) housing for higher pressures or aggressive fluids

All OpenSCAD files are parametric so these variants are straightforward to explore.

Design Comparison: HDD Platter vs CD/DVD Version

This document compares the two physical implementations of the Precision Flow Conditioner.

Summary Table

Aspect	HDD Platter Version	CD/DVD Version	Winner / Notes
Disc Diameter	95 mm	120 mm	CD/DVD (more flow area)
Disc Thickness	~0.9 mm	~1.2 mm	HDD (tighter gaps possible)
Surface Quality	Excellent (mirror polish)	Good (polycarbonate, varies by brand)	HDD clear winner
Cost / Availability	Medium (scrap Enterprise drives)	Very low (blank or used CDs/DVDs)	CD/DVD
Consistency	High (matched platters)	Medium (brand variation)	HDD
Central Hole	Small + custom	Standard 15 mm	CD/DVD easier
Max Stack Height	Excellent	Good	HDD
Best For	Highest performance / research	Prototyping, low budget, education	Depends on goal
Ease of Sourcing	Medium	Extremely easy	CD/DVD
Repeatability	Higher	Good (with good blank media)	HDD

Detailed Comparison

1. Flow Characteristics

HDD Version: Superior due to mirror-polished surfaces. Creates the cleanest boundary layers and lowest turbulence.

CD/DVD Version: Slightly higher surface roughness. Still very usable, especially with high-quality blank media. May produce marginally more turbulence at high flow rates.

Recommendation: Use HDD platters when you need the absolute best data quality. Use CD/DVD for development, teaching, or when budget/sourcing is the constraint.

2. Mechanical Construction

HDD platters are thinner → can fit more channels in the same height.

CD/DVD discs are thicker and have a large central hole → requires a different hub design (already provided in `cd_dvd_version/`).

3. Cost & Accessibility

This is the strongest argument for the CD/DVD version:

Blank CD-Rs or DVD-Rs can be bought for pennies each.

Many people already have stacks of old CDs/DVDs.

No need to hunt for specific Enterprise hard drives.

4. Experimental Use Cases

Choose HDD Platters when:

You are doing serious LENR/cavitation research and need maximum repeatability.

You already have access to matched platters.

You want the highest possible performance.

Choose CD/DVD when:

You are prototyping or teaching.

Budget is very limited.

You want to quickly build multiple units.

You are testing the overall concept before investing in premium platters.

Hybrid Recommendation

Many experimenters may want to do this:

1. Build the **CD/DVD version first** for learning, tuning, and initial experiments.
2. Once the experimental protocol is solid, build the **HDD platter version** for high-quality data collection.

The OpenSCAD code is parametric enough that switching between the two is relatively straightforward.

File Locations

HDD version: `scad/` (main files)

CD/DVD version: `scad/cd_dvd_version/`

Comparison notes are also embedded in the individual CD/DVD SCAD files.

Bottom line: The HDD version is higher performance. The CD/DVD version is dramatically more accessible. Both are valid and useful.

Theory, Design Rationale, Expected Function, and Value to MFMP Research

Why This Design? (Rationale from the Original Conversation)

The device was not conceived as another Tesla turbine for power generation. The goal was inverted:

Create a **precision, low-turbulence, multi-channel flow conditioner** that can deliver highly repeatable gas or liquid injection into a cavitation chamber.

Core Problems This Solves in Cavitation Experiments

Traditional injection methods (simple orifices, bubblers, or direct ultrasonic horn bubbling) suffer from:

High turbulence and chaotic bubble size distribution

Poor run-to-run and lab-to-lab repeatability

Difficulty systematically varying one parameter (velocity, swirl, shear, gas type) while holding others constant

Limited ability to pre-condition flow before it enters the high-energy cavitation zone

The HDD platter stack + central HVAT directly attacks these problems using well-understood fluid mechanics (boundary layer adhesion and organized spiral flow) in a form factor that is cheap, replicable, and scientifically useful.

Why Polished Enterprise HDD Platters?

Sub-nanometer mirror polish → excellent boundary layer attachment with minimal surface disruption.

Extremely consistent thickness and flatness when batch-selected.

Free/cheap from scrap server drives.

Pre-existing central geometry and mounting holes.

Proven in the DIY Tesla turbine community for low-Reynolds-number viscous work.

Why Stationary (or Very Lightly Rotating) Operation?

In classic Tesla turbines the discs spin and extract energy. Here the stack is primarily **stationary**. Its job is to force incoming tangential flow into a smooth, multi-channel inward spiral with controlled velocity decay and minimal chaotic mixing. This produces a conditioned, low-turbulence jet at the center.

Why the Central HVAT (Hybrid VAWT)?

The residual swirl leaving the platter stack still carries organized kinetic energy. The lightly loaded central rotor (Savonius starting scoops + helical Darrius blades) uses that energy to:

Further straighten the flow into a more axial, vortex-stabilized jet.

Act as a tunable mixer / flow straightener (via brake or light drive).

Dramatically improve the quality of the jet that then enters the cavitation nozzle or chamber.

It is **not** primarily a power extractor. It is an active fluid dynamic component.

Why This Specific Approach? Creating a Consistent Replication Testbed

The deeper goal of this project is not just to build “a better bubbler.” It is to help move the field of cavitation-based LENR and anomalous phenomena research toward **higher experimental rigor and true replicability across different labs and builders**.

The Core Problem in Current Cavitation Research

Many of the most interesting claims in this field (transmutations, excess heat, “strange radiation,” EVO formation, etc.) come from experiments that are extremely sensitive to small variations in bubble dynamics. When different researchers try to replicate each other, they often get different results — sometimes dramatically so.

Common injection methods (ultrasonic horns alone, simple gas bubblers, or direct liquid flow) have major weaknesses for replication work:

High variability in bubble size distribution from run to run and lab to lab.

Difficulty controlling or even measuring key parameters (local shear, vorticity, pre-existing microbubbles).

Strong dependence on operator technique and specific equipment quirks.

Poor ability to systematically vary one variable while holding others constant.

This makes it very hard to know whether different results come from real physical differences or just from uncontrolled differences in how the bubbles were created.

How This Device Is Intended to Help

The platter stack + HVAT approach was chosen specifically because it offers several structural advantages for creating a **more consistent, better-characterized injection platform**:

Multi-channel laminar-to-transitional flow: Instead of one chaotic jet or random bubbling, the device creates dozens of parallel, relatively well-behaved flow channels. This tends to average out some sources of randomness.

Controllable pre-conditioning: The stationary disc stack + adjustable HVAT brake lets the experimenter deliberately add or remove organized swirl before the fluid reaches the cavitation zone. This is a variable that is very difficult to control with simpler methods.

Scalable and modular: Changing the number of discs, gap size, nozzle angle, or HVAT brake setting gives relatively independent control knobs. This makes it easier (in principle) to run systematic parameter studies.

Visual access during characterization: The clear viewport option allows direct observation of the spiral flow pattern during low-pressure air testing — something that is usually impossible with sealed ultrasonic setups.

Low-cost replicability: Because the design uses either scrap hard drive platters or cheap CD/DVD media + 3D printing, it is realistic for many independent researchers to build very similar (or identical) versions of the same hardware.

The long-term vision is that if multiple labs build versions of this device (or improved descendants of it), they will at least be starting from a more similar fluid dynamic baseline than is common today. That would make it much more meaningful when different groups do — or do not — see the same anomalous effects.

This is why so much emphasis has been placed on:

Calibration pieces (so builders can actually know what their hardware is doing)

Example configurations with clear rationales

Detailed documentation of every design choice

The separation between the physical device and the speculative models

None of this guarantees success. It is simply an attempt to raise the quality floor of the experimental platform itself.

Why Not Just Use a Better Ultrasonic Horn or Commercial Cavitator?

Commercial equipment is often expensive, closed-box, and still has significant uncontrolled variables. Many researchers in this field operate on very limited budgets. The goal here was to create something that is:

Cheap enough for independent researchers

Open enough that people can understand and modify it

Characterizable enough that different builders can know they are doing something similar

Whether this particular geometry ultimately proves superior is something that only real experimental work by multiple independent groups can determine. This project is an attempt to make that experimental work more feasible and more comparable across labs.

Why the Tunable Venturi Cavitation Nozzle?

After the flow has been beautifully conditioned, we want to convert that organized energy into controlled cavitation at a known location. The converging-diverging nozzle + interchangeable orifice inserts give the experimenter direct control over:

Jet velocity

Pressure drop

Resulting bubble size distribution and collapse intensity

How It Is Expected to Function (Fluid Dynamics)

1. **Tangential Inlet** — Six nozzles at $\sim 8\text{--}12^\circ$ from true tangent introduce flow with strong tangential momentum into the gaps between platters.
2. **Boundary Layer Spiral** — Viscosity causes the fluid to adhere to the highly polished disc surfaces. Momentum is transferred inward in a controlled spiral (the classic Tesla effect, but here used for conditioning rather than torque).
3. **Multi-Channel Low-Turbulence Transport** — 24 parallel gaps (for 25 platters at 1 mm) act like a high-quality flow straightener bundle. Turbulence is damped; velocity profiles across the channels become similar.
4. **Central Collection + HVAT Conditioning** — Flow converges at the center. The HVAT rotor organizes residual swirl into a cleaner axial jet.
5. **Final Acceleration & Cavitation** — The Venturi nozzle accelerates the conditioned flow and creates the sharp pressure drop needed for reproducible bubble inception right at the injection point into the experimental chamber.

Key tunable parameters:

Inlet pressure / total flow rate

Nozzle angle and count (fixed in current design but parametric)

Inter-platter gap

HVAT brake setting (controls residual swirl)

Orifice diameter (controls final jet velocity and σ — cavitation number)

Expected Advantages for Reproducibility

Much tighter bubble size distribution than raw bubbling or simple orifices.

Ability to pre-load the chamber with microbubbles of known characteristics before applying ultrasound (hybrid ULTR mode).

Systematic variation of vorticity (via HVAT) while holding other variables.

Visual access (clear viewport) during air characterization phases.

Practical Operation: Setup, Tuning, Testing, and Tracking

A. Setup

1. Print all parts (see `build.md`).
2. Clean 25 matched platters with 99% IPA.
3. Assemble stack in the jig using 1.0 mm spacers.
4. Install HVAT on bearings (critical: free rotation with minimal play).
5. Mount on sturdy vertical stand above or beside your cavitation chamber.
6. Connect regulated gas supply (start with low-pressure air or argon) to the manifold.
7. Install chosen orifice in the cavitation nozzle.

B. Tuning

Basic characterization (air only): Vary inlet pressure while observing spiral through viewport. Note HVAT RPM with brake open vs. partially closed.

Jet quality: Adjust HVAT brake to find the setting that gives the cleanest, most axial exit jet (visual or with high-speed video).

Cavitation intensity: Swap orifice inserts. Smaller orifices = higher velocity = lower cavitation number (more violent collapse).

Hybrid mode: Introduce ultrasonic field and observe interaction between the pre-conditioned stream and the acoustic field.

C. Testing Protocol (Recommended Baseline)

See the detailed protocol in `testing.md`. High-level flow:

1. Low-pressure air visualization (mandatory first step).
2. Plain water + aluminum foil baseline (classic Greenyer ULTR signature).
3. Introduce conditioned gas stream.
4. Add ultrasound.
5. Systematic matrix (change one variable at a time: gas type, HVAT brake, orifice size, ultrasonic power).

D. Tracking Results

Strongly recommended minimum dataset per run:

Inlet pressure + total flow rate (or manifold pressure)

HVAT rotation speed (if measurable) or brake position

Orifice diameter used

Ultrasonic power/frequency

Gas type and any pre-saturation (argon, deuterium, etc.)

Before/after photos + video of foil

Temperature logs

Any radiation / neutron detector readings (with proper controls)

Post-run SEM/EDX or mass-spec on witnesses when available

The entire point of this device is to make the *fluid dynamic boundary conditions* far more repeatable than before, so that any anomalous effects can be more confidently attributed to the variables you are actually changing.

How This Is Unique and Adds Value to Bob Greenyer / MFMP Work

Bob Greenyer's research emphasizes **radical accessibility, radical documentation, and radical repeatability at low cost**. This device aligns perfectly:

Repeatability upgrade: Turns one of the most variable parts of cavitation experiments (injection) into a controlled, characterizable component.

New experimental axes: Ability to deliberately introduce controlled vorticity or pre-conditioned microbubbles before the ultrasonic field is applied.

Hybrid hydrodynamic + acoustic: Enables cleaner tests of the interaction between organized flow and ultrasound — an area Greenyer has highlighted as promising.

Open hardware contribution: Fully parametric, 3D-printable from scrap, documented with drawings and DXF profiles. Anyone in the community can build and improve it.

Data quality: Better boundary conditions → cleaner before/after data → stronger papers and replication attempts.

This is not “just another gadget.” It is a precision scientific instrument designed specifically to raise the quality floor of the kind of low-budget, high-insight cavitation work that the MFMP community has pioneered.

References: See `references.md` for the specific Greenyer videos, presentations, and forum threads that directly inspired the design goals and expected use cases.

Tools and Software

This project relies on a modern, fully open-source toolchain for parametric design, analysis, visualization, fabrication data generation, and documentation.

All tools mentioned here are free (as in freedom and usually as in price) and widely available.

Core Design & CAD

OpenSCAD — The primary tool for all parametric 3D models (`scad/` directory).

Every part is defined with variables at the top of each `.scad` file so you can easily customize dimensions, number of discs, nozzle angles, etc.

Highly recommended for anyone wanting to modify the design.

FreeCAD — Used for importing STL files and converting them to STEP format when needed for CNC quotation services (e.g., PCBWay).

Visualization, Analysis & Fabrication Data Generation

All the following Python tools live in the `python/` directory and were developed specifically for this project:

`generate_engineering_drawings.py` — Produces the full set of professional orthographic engineering drawings and 3D cutaway/isometric views (PNG + SVG).

`generate_dxf_exports.py` — Creates 2D DXF profiles with proper layers (CUT, DRILL, ANNOTATE) for laser cutting or CNC of flat parts (spacers, orifice inserts, bolt patterns, etc.).

`generate_flow_animation.py` — Generates the animated flow path visualization (individual frames + GIF).

`generate_performance_graphs.py` — Creates the sensitivity plots showing effects of HVAT brake position, orifice size, inlet pressure, and gap on performance metrics.

These scripts use:

NumPy — Numerical computations

Matplotlib — All 2D plots, 3D visualizations, and diagram generation

Pillow (PIL) — Image processing for animations and diagram post-processing

SciPy (where applicable) — Additional scientific computing

Documentation & Typesetting

Sphinx + MyST — This entire documentation site (`docs/`)

LaTeX (TeX Live) + natbib — The formal technical report (`latex/main.tex`)

Inkscape — SVG editing and final figure preparation

3D Printing & Slicing

See the dedicated [Printing Guide](#)..... 28 and the root [PRINTING.md](#) ↗ in the repository.

Profile templates are provided in `slicer_profiles/` for:

Bambu Studio / Orca Slicer (JSON)

PrusaSlicer (INI)

Cura (curaprofile)

How to Customize the Design

The project is built from the ground up for easy customization:

1. OpenSCAD Parametric Models (`scad/` and `scad/cd_dvd_version/`)

Every major file starts with a clear `// PARAMETERS` section. Examples:

`num_platters` or `num_discs`

`gap`

`nozzle_angle`

`hv_dia` (HVAT diameter)

Disc/platter diameter and thickness

Simply change the numbers at the top, render (F6), and export STL.

The CD/DVD version and HDD platter version are maintained in parallel so you can experiment with both.

2. Python Generators

Most generators accept parameters via command-line arguments or by editing the top of the script (e.g., changing default gap, number of frames in animation, pressure ranges in graphs, etc.).

Example:

```
python python/generate_performance_graphs.py
```

You can fork and modify any of these scripts to produce new plots, different DXF layers, or new diagram styles.

3. Calibration Pieces (`calibration_test_pieces/`)

Use these to dial in your specific printer + filament combination before committing to a full build:

Spacer thickness calibration

Bearing bore test rings

Nozzle throat test rings

Flatness test plate

4. DXF Exports (`dxf_exports/`)

These are generated from Python and can be further edited in Inkscape or imported directly into laser cutters / CNC software.

Replicating the Device with 3D Printing and Assembly

The entire design philosophy prioritizes **radical replicability** using only consumer-grade equipment.

3D Printing Strengths of This Project

All structural parts are designed to be printed on standard FDM printers (no resin required for the basic version).

Minimal or zero supports needed on most parts.

Critical tolerances (especially spacer thickness and bearing bores) are called out explicitly in the SCAD files and printing guides.

Calibration test pieces are provided so you can validate your printer before printing the real parts.

Assembly Advantages

The dedicated `ASSEMBLY_JIG.scad` ensures the disc stack is perfectly straight and gaps are uniform — this is one of the biggest reasons the device performs well.

Shipping inserts protect the precise gaps during transport.

Clear viewport option allows visual verification of flow during initial air testing.

Detailed instructions are in:

Build & Assembly Guide	32
Printing Guide.....	28
How to Print Precise Spacers	30

Navigation – Special Interest Areas

Use the following directory trees to quickly jump to areas of interest:

CAD & Parametric Design

```
scad/
├── HDD_Platter_Cavitation_Conditioner.scad  ← Main housing (HDD version)
├── HVAT_BLADES.scad
├── BEARING_HOUSERS.scad
├── CAVITATION_NOZZLE.scad
├── SPACER_RING.scad
├── ASSEMBLY_JIG.scad
├── cd_dvd_version/                               ← Complete alternative design using CD/DVD discs
│   ├── CD_DVD_Flow_Conditioner.scad
│   └── ...
```

Python Generators & Visualization

python/	
├─ generate_engineering_drawings.py	← Orthographic + cutaway views
├─ generate_dxf_exports.py	← Laser/CNC ready 2D profiles
├─ generate_flow_animation.py	← Flow development animation
└─ generate_performance_graphs.py	← Tuning sensitivity plots

Fabrication & Calibration

calibration_test_pieces/	← Printer calibration tools
dxf_exports/	← Ready-to-use DXF files
licer_profiles/	← Bambu, Prusa, Cura profiles
fabrication/	← PCBWay quote packages

Documentation

docs/	
├─ printing_guide.md	
├─ build.md	
├─ tools_and_software.md	← This page
├─ engineering_drawings.md	
├─ gallery.md	
└─ theory_and_value.md	

See the root [PROJECT_STRUCTURE.md](#) ↗ for the complete tree.

All of these tools and files are deliberately chosen or created so that a motivated individual or small group can replicate, understand, modify, and improve the device using only free software and common 3D printing hardware.

Customization Guide

This guide shows exactly how to modify the design for different experimental goals using the provided OpenSCAD models and Python tools.

All changes are parametric — no need to rewrite geometry from scratch.

1. OpenSCAD Parameter Customization (Easiest Starting Point)

HDD Platter Version (`scad/HDD_Platter_Cavitation_Conditioner.scad`)

Open the file and edit the top section:

```
// ===== PARAMETERS (tweak these) =====
platter_od = 95;           // mm – actual Enterprise 3.5" platter diameter
num_platters = 25;        // change this for taller/shorter stack
platter_thick = 0.9;       // typical enterprise ~0.9 mm
gap = 1.0;                 // inter-platter gap (0.8–1.5 mm sweet range)

nozzle_count = 6;         // 4–8 recommended
nozzle_angle = 10;        // degrees from true tangential (8–12° optimal)
nozzle_w = 5.5;           // slit width at exit

hv_dia = 28;              // central HVAT rotor diameter
diffuser_len = 38;
diffuser_outlet_dia = 44;
```

Common modifications and expected effects:

Increase `num_platters` to 30–35: More flow channels, potentially smoother output, taller device.

Decrease `gap` to 0.8 mm: Higher velocity and shear for same flow rate (more intense cavitation possible, harder to print consistently).

Change `nozzle_angle` to 8°: Stronger inward spiral, potentially better conditioning but higher pressure drop.

Change `nozzle_angle` to 15°: Easier flow, less aggressive spiral.

After editing, press **F6** (Render), then **File** → **Export** → **Export as STL**.

CD/DVD Version (`scad/cd_dvd_version/CD_DVD_Flow_Conditioner.scad`)

Similar parameters, but tuned for 120 mm discs:

```
disc_od = 120;
disc_thickness = 1.2;
num_discs = 20;           // Often fewer discs than HDD version because discs are thicker
gap = 1.2;                 // Slightly larger recommended gap

nozzle_angle = 10;
hv_dia = 32;              // Larger central hub for CD/DVD 15mm hole
```

Tip: The CD/DVD version is excellent for rapid prototyping different gap/nozzle configurations because blank discs are cheap and easy to source.

2. Python Generator Customization Examples

Generate Performance Graphs for a Specific Configuration

Edit `python/generate_performance_graphs.py` or run it after modifying the internal models.

Example — to explore effect of very small gaps:

```
# In generate_performance_graphs.py, modify the gap range
gaps = np.linspace(0.5, 1.5, 50) # was 0.6 to 1.8
```

Then run:

```
python python/generate_performance_graphs.py
```

New graphs will be saved to `illustrations/performance_graphs/`.

Generate Engineering Drawings for Custom Parameters

The drawing generator currently uses hardcoded defaults that match the OpenSCAD defaults.

To match a custom build:

1. Edit the top of `python/generate_engineering_drawings.py` (platter_od, num_platters, gap, nozzle_angle, etc.).
2. Run:

```
python python/generate_engineering_drawings.py
```

This will produce new PNG/SVG files reflecting your exact configuration.

Generate DXF Files for Custom Disc Diameter

Example: You built the CD/DVD version with 20 discs at 1.3 mm gap:

Edit `python/generate_dxf_exports.py` and change:

```
write_spacer_ring_dxf("my_custom_spacer.dxf", od=120, id=16, thickness=1.3)
```

Then run the script. The resulting DXF can be used directly for laser cutting alternative spacers (e.g., from thin plastic sheet or metal).

4. Before / After OpenSCAD Examples for Common Modifications

Here are practical before-and-after diffs for the most common useful changes.

Example A: Switch from Default 25 platters @ 1.0 mm gap to a taller, gentler stack (32 platters @ 1.4 mm gap)

Before (default):

```
num_platters = 25;
gap = 1.0;
```

After:

```
num_platters = 32;    // +7 discs → more parallel channels
gap = 1.4;            // gentler flow, easier printing
```

Expected outcome: Smoother time-averaged flow, lower shear per channel, easier to manufacture spacers.

Example B: More aggressive cavitation (smaller nozzle angle + smaller orifice)

Before:

```
nozzle_angle = 10;
```

After (in the cavitation nozzle file as well):

```
nozzle_angle = 7;    // stronger spiral
```

Then in your cavitation nozzle usage:

```
cavitation_jet_nozzle(..., throat_d=6);    // instead of 8 mm
```

Expected: Higher jet velocity and lower cavitation number → more violent bubble collapse.

Example C: Maximize HVAT conditioning effect (larger rotor for CD/DVD version)

In `scad/cd_dvd_version/CD_DVD_Flow_Conditioner.scad`:

Before:

```
hv_dia = 32;
```

After:

```
hv_dia = 36;    // larger rotor for stronger swirl conditioning
```

Remember to also adjust the corresponding hub and bearing houser files in the same folder.

Example D: Quick prototyping with CD/DVD version (fewer discs)

In `scad/cd_dvd_version/CD_DVD_Flow_Conditioner.scad`:

Before:

```
num_discs = 20;
```

After (for very fast iteration):

```
num_discs = 12;    // shorter stack, faster prints, good for initial tuning
```

This is excellent when you want to test multiple gap or angle combinations quickly.

Pro tip: Keep your “golden” working parameters in a separate text file or commented block at the top of each SCAD file so you can easily switch between “research mode” and “rapid prototype mode”.

3. Visual “Customization Quick Reference” Diagram

A diagram has been generated showing the main control knobs and their primary effects:

Location:

`illustrations/performance_graphs/customization_quick_reference.png` (and .svg)
Printable one-page A4 PDF:
`illustrations/performance_graphs/Customization_Quick_Reference_Printable.pdf` (also in
`docs/_static/`)

It is also embedded in the Sphinx documentation under the Tools and Software section for easy navigation.

The diagram illustrates the relationship between:

Gap size → Reynolds number & flow regime

Nozzle angle → Spiral strength vs. pressure drop

HVAT brake position → Swirl reduction vs. jet coherence

Number of discs → Number of parallel channels vs. stack height

Orifice diameter → Jet velocity & cavitation intensity

(If the file does not exist yet in your checkout, run `python python/generate_performance_graphs.py` after the latest updates.)

4. Simulated Performance Characteristics (Important Disclaimers)

The graphs in `illustrations/performance_graphs/` are **software estimates only**, based on simplified analytical models (continuity + boundary layer scaling + cavitation number definitions).

They are NOT CFD results and should be treated as directional guidance only.

What Different Configurations Are Expected to Do (Qualitative)

Smaller gaps (0.6–0.8 mm):

Higher local velocities for the same mass flow

Higher Reynolds numbers → transition toward more turbulent (but still organized) flow

Potentially more intense shear and smaller initial bubble sizes

Harder to manufacture consistently

Larger gaps (1.4–1.8 mm):

Lower velocities, gentler cavitation

Easier printing and assembly

Possibly larger, more chaotic bubbles

Lower nozzle angle (6–8°):

Longer spiral path inside the stack

More opportunity for boundary layer momentum transfer

Higher pressure drop across the device

Higher nozzle angle (12–15°):

Shorter, more direct flow path

Easier to drive high flow rates

Less conditioning effect

HVAT brake partially engaged (30–50% locked):

Significant reduction in residual swirl

Much cleaner, more axial jet entering the cavitation chamber

One of the highest-leverage “tuning knobs” available without reprinting parts

More discs (30+):

More parallel flow channels

Potentially smoother time-averaged output

Taller overall device, more material usage

These behaviors are consistent with boundary layer theory and observed behavior in similar devices, but every physical build will deviate based on surface finish, alignment precision, inlet manifold quality, etc.

Recommendation: Use the Python generators to explore “what if” scenarios, then validate with your physical prototype using the calibration pieces and viewport.

Next Steps for Experimenters

1. Start with the default parameters in both the HDD and CD/DVD versions.
2. Use the calibration pieces to lock in your printer’s real-world tolerances.
3. Build one device with the defaults.
4. Use the Python tools to predict the effect of one change at a time.
5. Modify only that parameter, rebuild, and compare results.

Ready-to-Use Example Configurations

For convenience, several pre-tuned configurations are provided in the `configs/` directory:

`aggressive_cavitation/` — Maximize collapse intensity

`gentle_high_throughput/` — High flow with milder conditions

`maximum_uniformity/` — Best conditioning quality

`rapid_prototyping_cd_dvd/` — Fast iteration using cheap discs

Each folder contains a modified `.scad` file + a README explaining the rationale and expected behavior. Copy the file into your working directory to use it.

This systematic approach is exactly what makes the device valuable for serious cavitation research.

All source files are designed to make this kind of iterative experimentation as frictionless as possible.

Software Bill of Materials (SBOM)

This document lists all major software components used to design, analyze, visualize, and document the Precision Flow Conditioner project.

Design & CAD

OpenSCAD (2021.01 or newer) — Primary parametric modeling tool for all 3D parts.

FreeCAD — Used for STL → STEP conversion when preparing files for CNC quotation services.

Analysis & Visualization

Python 3 (3.10+)

NumPy — Numerical computations for flow models and performance graphs

Matplotlib — All engineering drawings, 3D cutaways, plots, and diagram generation

Pillow (PIL) — Image processing for flow animation frames

SciPy (optional) — Additional scientific utilities

Documentation & Typesetting

Sphinx + **MyST Parser** + **Furo** theme — This documentation website

LaTeX (TeX Live 2026) + **natbib** — Formal technical report (`latex/main.tex`)

Inkscape — Final SVG editing and figure preparation

3D Printing & Fabrication Data

Bambu Studio / **Orca Slicer** / **PrusaSlicer** / **Cura** — Slicing (profile templates provided in `slicer_profiles/`)

Custom Python generators (see `python/`) for DXF laser/CNC files and performance data

Version Control & Collaboration

Git

GitHub (or equivalent)

All of the above are free and open source software.

This SBOM is intended to help replicators understand the complete digital toolchain required to work with or extend the project.

Comprehensive 3D Printing Guide

This is the definitive guide for successfully printing all parts of the HDD Platter Precision Flow Conditioner.

Material Recommendations

Material	Recommendation	Pros	Cons	Notes
PETG	Best overall choice	Excellent chemical resistance, good strength, low warping, easy to print	Can be stringy	Primary recommendation for most users
ASA	Excellent alternative	UV resistant, very good chemical resistance, tough	Slightly harder to print than PETG	Great if device will see sunlight or chemicals
ABS	Acceptable	Good strength, widely available	Warps more, needs enclosure	Only if you have good ABS printing setup
PLA	Not recommended	Easy to print	Brittle, poor chemical/heat resistance	Will fail in real use
Nylon	Advanced	Very tough	Hygroscopic, difficult	Only for experienced users
Polycarbonate	Advanced	Extremely tough	Very difficult to print	Overkill for most builds

Winner for most people: PETG

Per-Part Printing Recommendations

High Priority / High Tolerance Parts

Part	Material	Layer Height	Infill	Perimeters	Special Notes
Spacer Rings (24+)	PETG	0.12-0.15 mm	80-100%	4-5	Most critical part. Print all at once from same spool. Measure thickness.
Bearing Housers	PETG/ASA	0.15 mm	50%+	6+	Bore tolerance is everything. Test print first.
HVAT Blades	PETG/ASA	0.15-0.2 mm	40-50%	5+	High strength needed. Balance after printing.
Main Housing	PETG	0.2 mm	30-40%	4-5	Large part — good first layer + brim essential.
Cavitation Nozzle	PETG/ASA	0.15-0.2 mm	40%	4	Sand internal surfaces for best flow.

Lower Priority Parts

Assembly Jig: 0.2mm, 35% infill, 5 perimeters (needs to stay flat)

Clear Viewport: 0.2mm, 30% infill (use clear PETG if possible)
Shipping Inserts: 0.2mm, 30% infill (thicker = more protection)
Operation Covers: 0.2mm, 30-35% infill

Slicer-Specific Recommendations

Bambu Studio / Orca Slicer (Recommended)

Use **0.2mm Standard** profile as base
Increase **Wall loops** to 4-5 for structural parts
Set **Infill** to Gyroid 35%+ for most parts
Enable **Arachne** wall generator
Use **Thick bridges** for the large housing

PrusaSlicer

Use **0.2mm QUALITY** or **0.15mm QUALITY**
Increase **Perimeters** manually
Use **Gyroid** infill
Enable **Elephant foot compensation** (0.2-0.3mm) for spacers

Cura

Use **Standard Quality** or create custom 0.15-0.2mm profile
Increase **Wall Line Count** to 5
Use **Gyroid** or **Cubic** infill
Enable **Fuzzy Skin** on non-critical external surfaces (optional)

Critical Tolerances & How to Achieve Them

1. Spacer Thickness (± 0.05 mm target)

Print at very fine layers (0.12-0.15mm)
Print everything in one job from the same filament
Measure 5 random spacers after printing
Adjust flow rate if consistently off

2. Bearing Bore (608ZZ)

Target: 8.15 – 8.25 mm
Print a test ring first with different sizes
Vertical printing of the bore gives best roundness

3. Nozzle Throat Diameter

Critical for experimental repeatability
Print orifices at finest layer height possible
Consider printing them solid (100% infill)

Common Problems & Solutions

Problem	Likely Cause	Solution
Spacers too thick	Elephant's foot + high flow	Increase first layer height, reduce flow 3-5%
Bearing too loose	Wrong tolerance in model or poor cooling	Print with smaller bore test, improve cooling
Housing warps	Poor bed adhesion / cooling	Use brim, enclosure for ABS, good first layer
Poor surface on internal flow paths	Low perimeters	Increase perimeters to 5+
HVAT doesn't spin freely	Bearing bore issues or debris	Ream/sand bore, clean thoroughly

Post-Processing Recommendations

Annealing PETG (optional but recommended for chemical resistance): 80°C for 1 hour in oven

Light sanding on mating surfaces and internal flow paths

Thorough cleaning of all holes and threads

For orifices: Consider drilling to final size if you have the bits

See Also

`docs/how_to_print_precise_spacers.md` — Detailed guide for hitting the most critical tolerance in the project (± 0.05 mm on spacers)

Recommended Workflow

1. Print 5-6 test spacers first → measure and dial in flow
2. Print a test bearing housing ring → verify bore fit
3. Print the Assembly Jig + a few spacers
4. Print the HVAT blades (high strength)
5. Print the main housing (biggest time investment)
6. Print remaining small parts

This order lets you catch problems early.

For the absolute latest recommendations and community results, check the GitHub issues and discussions once the project is public.

How to Hit ± 0.05 mm Spacer Thickness (Detailed Guide)

The spacer rings are the most critical printed parts for device performance. Inconsistent gaps destroy the low-turbulence flow you're trying to achieve.

Target: ± 0.05 mm (ideally ± 0.03 mm) on 1.0 mm or 1.3 mm spacers.

Step-by-Step Calibration Process

1. Print the Calibration Set

Use the file: `calibration_test_pieces/Spacer_Thickness_Calibration.scad`

Print 6–8 pieces in one job using your final intended settings.

2. Measure Properly

Use a good digital caliper (0.01 mm resolution)

Measure at center and at two points on the edge of each spacer

Record all measurements

Calculate average and standard deviation

3. Common Adjustments by Brand

Bambu Lab (X1/P1/A1) + Bambu Studio / Orca

Too thick overall: Reduce “Flow rate” by 2–4% (try 96–98%)

Elephant’s foot on first layer: Increase “Initial Layer Height” to 0.22–0.25 mm

Inconsistent thickness: Slow down “Top surface speed” to 80–100 mm/s

Best profile base: Use the provided `PETG_High_Precision_for_Device.json`

Prusa MK4 / Mini + PrusaSlicer

Too thick: Lower “Extrusion multiplier” to 0.96–0.98

First layer elephant’s foot: Enable “Elephant foot compensation” at 0.2–0.3 mm

Use 0.15 mm or 0.12 mm QUALITY profile as base instead of 0.20 mm

Creality / Generic Printers + Cura or Orca

Too thick: Reduce “Flow” to 96–98%

Inconsistent layers: Increase “Wall Line Count” and slow down top layers

Consider switching to a 0.15 mm or 0.12 mm profile for spacers specifically

4. Pro Tips for ± 0.03 mm Results

Print spacers **last**, after you’ve dialed in your printer on the big parts.

Use the **same filament spool** for the entire spacer batch.

Dry your filament (PETG especially loves moisture).

Print with a **0.12 mm or 0.15 mm layer height** for spacers even if you use 0.2 mm everywhere else.

Avoid “fuzzy skin” or any surface modifiers on spacers.

After printing, let parts cool completely before measuring.

5. When to Give Up on a Batch

If after 2–3 calibration rounds your standard deviation is still > 0.08 mm, consider:

Switching filament brands

Printing solid (100% infill) spacers instead of using lightening holes

Using a different material (some PETG brands are simply more consistent than others)

Quick Decision Tree

Average thickness $>$ target + 0.08 mm $\rightarrow$ Reduce flow

First layer much thicker than rest $\rightarrow$ Elephant foot compensation or first layer height

High variation between pieces $\rightarrow$ Environmental (temperature, moisture, speed)

Good luck — consistent spacers are what separate a good device from a great one.

Build & Assembly Guide

This guide walks you through printing, preparing, and assembling the 25-platter HDD Platter Precision Flow Conditioner exactly as developed in the May 2026 design conversation.

1. Print All Parts

$\rightarrow$ **See the comprehensive guide:** [docs/printing_guide.md](#).....28

The short version:

Best material: PETG (ASA is excellent alternative)

Use the separate OpenSCAD modules in `scad/`

Layer height: 0.2 mm (0.12–0.15 mm for spacers and bearing bores)

Infill: 30–50% (higher on HVAT blades and bearing housers)

Supports: Not required on these designs

Important: Print a few test spacers and a bearing bore test ring **before** doing the full job. Spacer thickness and bearing bore are the most critical dimensions.

Why This Design is Highly Replicable with Consumer 3D Printing

Unlike many scientific instruments, this device was deliberately engineered for radical accessibility:

All structural parts are designed for standard FDM printers (no industrial equipment required for the basic version).

The `ASSEMBLY_JIG.scad` solves one of the hardest problems in stacked-disc devices: keeping gaps perfectly uniform and the stack straight.

Shipping inserts protect the precise gaps during transport.

Clear viewport option lets you visually verify the beautiful spiral flow during initial low-pressure air testing — extremely useful for debugging and education.

Calibration pieces are provided so you can validate your specific printer + filament combination before printing the real parts.

Two parallel designs exist: the high-performance HDD platter version and the very low-cost CD/DVD version. See the [Printing Guide](#)²⁸ and [Tools and Software](#)¹⁸ pages for much deeper information on materials, settings, and customization.

Required prints (default 25-platter config):

1 × Main housing + nozzles + diffuser (`HDD_Platter_Cavitation_Conditioner.scad`)
1 × HVAT rotor (`HVAT_BLADES.scad`)
2 × Bearing housers (`BEARING_HOUSERS.scad`)
24 × Spacer rings (1.0 mm) (`SPACER_RING.scad`)
26 × Shipping inserts (1.4 mm) (`SHIPPING_INSERTS.scad`)
1 × Assembly jig (`ASSEMBLY_JIG.scad`)
1 × Clear viewport (optional but highly recommended for testing) (`CLEAR_VIEWPORT.scad`)
1 × Solid top cover + 1 × Bottom jet cover (`OPERATION_COVERS.scad`)
1 × Cavitation jet nozzle body + 4+ orifice inserts (6/8/10/12 mm) (`CAVITATION_NOZZLE.scad`)
1 × Nozzle manifold (distributes compressed air to the 6 tangential inlets)
4 × Sensor mounts (for pressure transducers)
HVAT brake disc + caliper (for tunable swirl)
Print a few extra spacers and inserts — they are small and easy to lose.

2. Prepare the Platters

1. Source 25 matched Enterprise 3.5" HDD platters (95 mm OD).
2. Clean **thoroughly** on both sides with 99% isopropyl alcohol. Any fingerprint or residue will create turbulence.
3. Inspect for warp or damage. Replace any suspect platters.

3. Assemble the Platter Stack (Use the Jig)

1. Place the Assembly Jig on a flat surface.
2. Insert the bottom end clamp plate.
3. Alternate: platter → spacer ring → platter (24 spacers for 25 platters).
4. Finish with the top end clamp plate (or clear viewport for testing).
5. Insert 6–8 long M3/M4 screws through the aligned holes and tighten evenly in a star pattern (1–1.5 Nm). Do not overtighten.

The jig keeps everything perfectly straight and the gaps uniform.

4. Install into Housing + HVAT

1. Carefully slide the clamped stack into the main housing (still in the jig if helpful).
2. Align the tangential nozzle openings with the inter-platter gaps.
3. Secure top and bottom end plates fully.
4. Press lower 608ZZ bearing into lower bearing houser.
5. Screw lower bearing houser to bottom end plate.
6. Insert central shaft + HVAT rotor (center it in the stack).
7. Press upper bearing into upper houser and mount to top plate.

8. Add thin thrust washer if needed.
9. Mount brake disc + caliper on top of shaft (for swirl control).

5. Add Flow & Sensor Features

Attach the 6-inlet nozzle manifold to the tangential nozzles (press-fit or screws + silicone tubing).
Install the 4 sensor mounts over the side ports (MPX5700 or similar pressure sensors recommended).
Screw or clamp the cavitation jet nozzle (with chosen orifice insert) to the bottom of the diffuser.
Add top/bottom covers as needed.

6. Final Checks

Rotate HVAT by hand — it should spin freely with minimal play.
Verify all gaps are uniform by looking through the clear viewport (or by feel with a feeler gauge).
For shipping: swap to thicker shipping inserts + solid covers.
The entire unit is now ready for low-pressure air testing and subsequent liquid cavitation experiments.
See `testing.md` for the recommended Greenyer-style protocol.

Operation: Setup, Tuning, Testing, and Result Tracking

This page provides concrete, step-by-step guidance for using the HDD Platter Flow Conditioner in real experiments.

1. Initial Setup (First Build)

Mechanical Assembly Checklist

☐

All 25 platters thoroughly cleaned with 99% IPA on both faces.

☐

Stack assembled in the jig using 1.0 mm spacers (24 pcs).

☐

HVAT rotor spins freely on bearings with very low friction (critical).

☐

Bearing housers securely mounted to end plates.

☐

Nozzle manifold attached and leak-free at low pressure.

☐

Clear viewport installed for first characterization runs.

First Air Test (Mandatory)

Connect low-pressure regulated air (start at 0.1–0.3 bar).

Observe through the viewport:

Is the inward spiral symmetric and beautiful?

Are all channels participating?

How does the HVAT behave with brake fully open vs. lightly applied?

Record video. This footage is scientifically valuable on its own.

2. Tuning Methodology

Primary Control Knobs (in rough order of impact)

Knob	Effect	Typical Range	Notes
Inlet pressure / flow	Overall velocity and Re in gaps	0.1 – 1.5 bar (air)	Start low
HVAT brake position	Residual swirl strength & jet coherence	Fully open → fully locked	Most powerful tuning parameter
Orifice diameter	Final jet velocity & cavitation number σ	6, 8, 10, 12 mm	Smaller = more intense
Inter-platter gap	Channel count vs. velocity (build-time)	0.8 – 1.5 mm	Fixed after assembly
Gas type	Density, viscosity, dissolved gas effects	Air, Ar, D ₂ , etc.	Major variable for LENR work

Recommended Tuning Sequence

1. Fix orifice at 8 mm.
2. Characterize spiral + HVAT behavior across a pressure sweep with brake open.
3. Find the brake setting that gives the subjectively “cleanest” exit jet (least turbulence, best axial coherence).
4. Lock that brake position (or note the setting) and begin liquid tests.
5. Later, deliberately vary brake position as an experimental axis.

3. Testing Protocols

Level 1: Air Characterization (Do This First)

Multiple pressures

Brake open / 50% / closed

High-speed or 4K video of spiral and exit jet

Optional: smoke or fog for flow visualization

Level 2: Basic Hydrodynamic Cavitation

Distilled water + aluminum foil witness

Conditioned air or argon injection

Systematic orifice swap

Document foil changes vs. injection parameters

Level 3: Hybrid ULTR (Highest Value)

Combine the platter device with 40 kHz ultrasound.

Key matrix variables:

With vs. without pre-conditioned gas stream

Different HVAT brake settings

Argon vs. air vs. deuterium-enriched

Ultrasonic power levels

This is where the device is expected to show its greatest advantage over simpler injection methods.

4. Result Tracking (Minimum Viable Dataset)

For every meaningful run, record at minimum:

Fluid dynamic boundary conditions

Inlet pressure (or mass flow if available)

HVAT state (brake position or measured RPM)

Orifice diameter

Gas type + any pre-treatment

Acoustic / Energy input

Ultrasonic power and frequency

Transducer mounting location

Environmental

Starting liquid temperature and volume

Ambient conditions

Observations

Video (through viewport + chamber)

Still photos of foil before/after

Temperature rise during run

Any visible light emission

Gas evolution notes

Post-run analysis (when possible)

SEM/EDX on foil

Mass spectrometry on evolved gas

Radiation detector logs (with controls)

The device's purpose is to make the first category ("Fluid dynamic boundary conditions") dramatically more repeatable and characterizable than in previous work.

5. Safety & Best Practices

Always start with air and plain water.

Use proper pressure regulation and relief.

Deuterium work: treat as flammable gas; use small quantities and good ventilation.

When using ultrasound + high voltage drivers, follow electrical safety practices.

If pursuing radiation claims, use appropriate detectors and conservative interpretation.

6. Iteration Ideas

Once you have baseline data, consider:

Variable-pitch or servo-controlled HVAT (future OpenSCAD module)

Integrated pressure sensors on the printed mounts

Different platter counts or gaps for throughput vs. conditioning trade-offs

Metal housing version for higher pressures or corrosive fluids

Direct coupling of the cavitation nozzle into a sealed chamber with optical access

Document everything and share it. That is how this tool becomes genuinely valuable to the broader research effort.

See also:

`build.md` for mechanical assembly details

`testing.md` for the detailed Greenyer-style protocol

`theory_and_value.md` for the scientific rationale

Testing Protocol (Greenyer / MFMP Style)

This protocol is derived directly from the design conversation and Bob Greenyer's public ULTR and cavitation work. It emphasizes radical documentation, low-cost repeatability, and safety.

Phase 1: Low-Pressure Air Characterization (Mandatory First Step)

Purpose: Verify beautiful, uniform inward spiral flow and HVAT behavior before introducing liquid.

Connect regulated compressed air (0.1 – 1.0 bar) to the nozzle manifold.

Observe and video the spiral through the clear viewport at multiple flow rates.

Record:

Inlet pressure

Approximate exit velocity (anemometer or high-speed video)

HVAT rotation speed (tachometer or video) with brake fully open vs. lightly applied

Look for any asymmetry, turbulence, or whistling (indicates a bad gap or nozzle alignment).

Photograph/video the exit jet with and without HVAT spinning.

This phase alone is extremely valuable for understanding the device and produces beautiful footage.

Phase 2: Basic Hydrodynamic Cavitation (Water + Foil)

1. Mount the device vertically above (or beside) a simple transparent cavitation chamber.
2. Use the 8 mm or 10 mm orifice insert initially.
3. Fill chamber with distilled water + small aluminum foil witness sample (classic Greenyer ULTR signature test).
4. Start at very low flow (just enough to produce visible bubbles).
5. Gradually increase flow / pressure while observing bubble cloud behavior through the chamber walls or view-port.

6. Run for 5–30 minutes. Document:

Temperature rise (or lack thereof)

Visible pitting/erosion patterns on foil (“snakes”, craters, microspheres)

Gas evolution

Any light emission (sonoluminescence) in dark conditions

Phase 3: Hybrid ULTR (Ultrasonic + Conditioned Injection)

This is the most promising configuration according to the conversation.

Add a 40 kHz, 50–100 W piezoelectric transducer (cheap ultrasonic cleaner driver + horn, or dedicated lab sonicator) to the bottom or side of the chamber.

Pre-saturate water with argon (brighter sonoluminescence) or use deuterium-enriched water for LENR-leaning runs.

Use the HVAT brake to tune residual swirl — many experimenters report that controlled vorticity helps stabilize certain effects.

Run controlled matrix:

With/without platter device (baseline comparison)

Different orifice sizes

HVAT braked vs. free vs. driven

Different dissolved gases

Ultrasonic power levels

Phase 4: Advanced / LENR-Interest Runs (with caution)

Deuterium gas sparging or D₂O

Witness materials: Pd, Ti, Cu, Ni foils or powders (post-run SEM/EDX or mass-spec)

Neutron / gamma monitoring + proper shielding

Long-duration runs with data logging (pressure, temperature, radiation)

Documentation Requirements (MFMP Spirit)

High-speed or 4K video of bubble behavior and foil changes

Before/after photos of every witness

Raw pressure, temperature, and radiation logs (timestamped)

Exact parameters (flow rate or pressure, HVAT RPM if measured, ultrasonic power, gas type, orifice size, gap setting)

SEM/EDX or mass-spec results when available

Share everything publicly on LENR-Forum, MFMP channels, or your own site. This device was built to make replication easier and data higher quality.

Safety Notes

Compressed air or gas cylinders: proper regulators and burst protection.

Deuterium: flammable — use in small quantities with good ventilation.

Ultrasonic + high voltage drivers: electrical safety.

If pursuing radiation claims: use appropriate detectors and shielding. Treat any positive results with extreme skepticism until independently replicated.

See `references.md` for the exact Greenyer videos and threads that inspired this protocol.

Ultrasonic Hybrid (ULTR-Style) Integration

The platter flow conditioner was explicitly designed to be paired with ultrasonic cavitation in the style of Bob Greenyer's well-known ULTR experiments.

Why Hybridize?

Pure ultrasonic or pure hydrodynamic cavitation each have strengths and weaknesses. Combining a precision-controlled hydrodynamic jet (from the HDD platter stack + HVAT) with a 40 kHz ultrasonic field gives experimenters the best of both:

Highly repeatable bubble size distribution and injection conditions (from the platter device)

Violent collapse energy from the ultrasonic field

Ability to tune vorticity (via HVAT brake) while the ultrasonic horn provides the main power

Greenyer and others have reported that certain anomalous effects (microspheres, transmutation signatures, "strange radiation") appear more reliably or intensely in well-controlled hybrid setups.

Recommended Transducer Setup

Frequency: 40 kHz (sweet spot for strong cavitation in water without excessive heating)

Power: 50–100 W for small chambers (2–5 L). Larger chambers need proportionally more.

Driver: Cheap generic ultrasonic cleaner boards on AliExpress/eBay work surprisingly well for initial experiments. For more control, use a dedicated laboratory ultrasonic processor with horn.

Mounting:

Bolt or epoxy the transducer/horn to the bottom of the cavitation chamber (most common).

Or immerse the nozzle exit directly into a small ultrasonic bath.

Use the clear viewport on the platter device to observe how the conditioned gas stream interacts with the ultrasonic field.

Expected Observable Signatures (from Greenyer's Work)

When running the hybrid system you should look for:

Classic foil erosion: "snakes", craters, pitting

Microsphere formation (often iron-rich and crenelated — a Greenyer signature)

Gas evolution (H₂ and others)

Possible light flashes (sonoluminescence) in darkened conditions

Temperature anomalies or excess heat (monitor carefully)

Transmutation traces on witness foils (requires SEM/EDX or mass spectrometry after the run)

"Strange radiation" tracks on CR-39 or other detectors (advanced / controversial)

Tuning Parameters

Record these for every run:

Inlet pressure / total flow rate through the platter device

HVAT rotation speed and brake setting (this directly affects swirl and bubble clustering)

Ultrasonic power and frequency

Gas type (air, argon for brighter SL, deuterium-enriched for fusion-leaning claims)

Orifice diameter (6/8/10/12 mm)

Witness material and placement

Chamber geometry and liquid volume

The combination of the platter device's low-turbulence spiral + tunable HVAT + ultrasonic field gives you far more knobs than a simple ultrasonic cleaner alone.

Practical Tips

Start with plain water + aluminum foil + 8 mm orifice + moderate ultrasonic power. Get the baseline “ULTR look” working first.

Then introduce the conditioned gas stream and observe how the bubble cloud changes.

Many researchers report that pre-existing microbubbles from the hydrodynamic jet seed the ultrasonic cavitation and change the collapse dynamics.

See the exact video links and data in `references.md`.

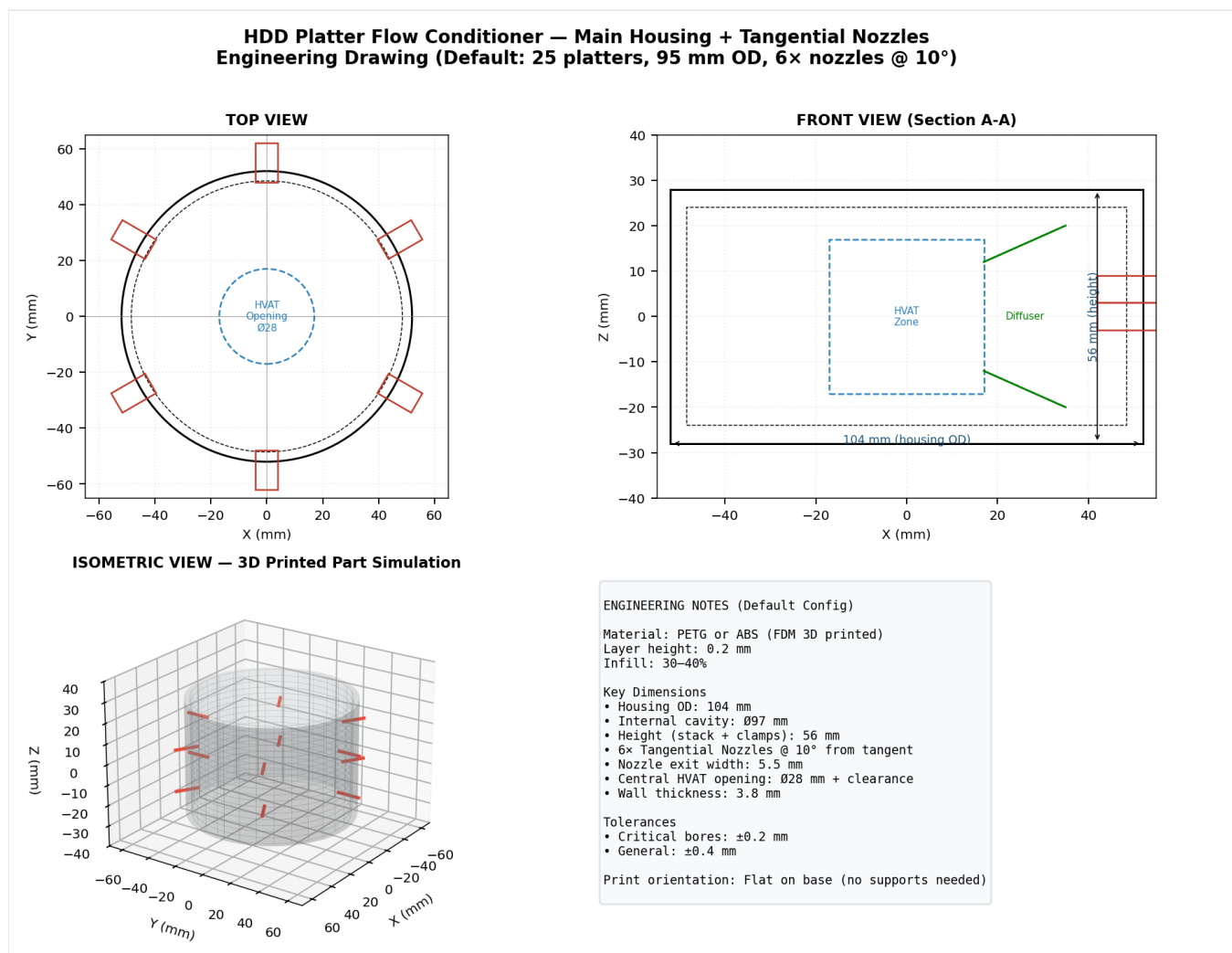
This hybrid approach is one of the most promising low-cost, high-documentation directions in current open LENR/cavitation research.

Engineering Drawings & Part Illustrations

This page provides professional engineering drawings and visual simulations of every major 3D-printable component in the HDD Platter Precision Flow Conditioner.

All drawings are generated programmatically from the parametric dimensions defined in the OpenSCAD files (default 25-platter configuration).

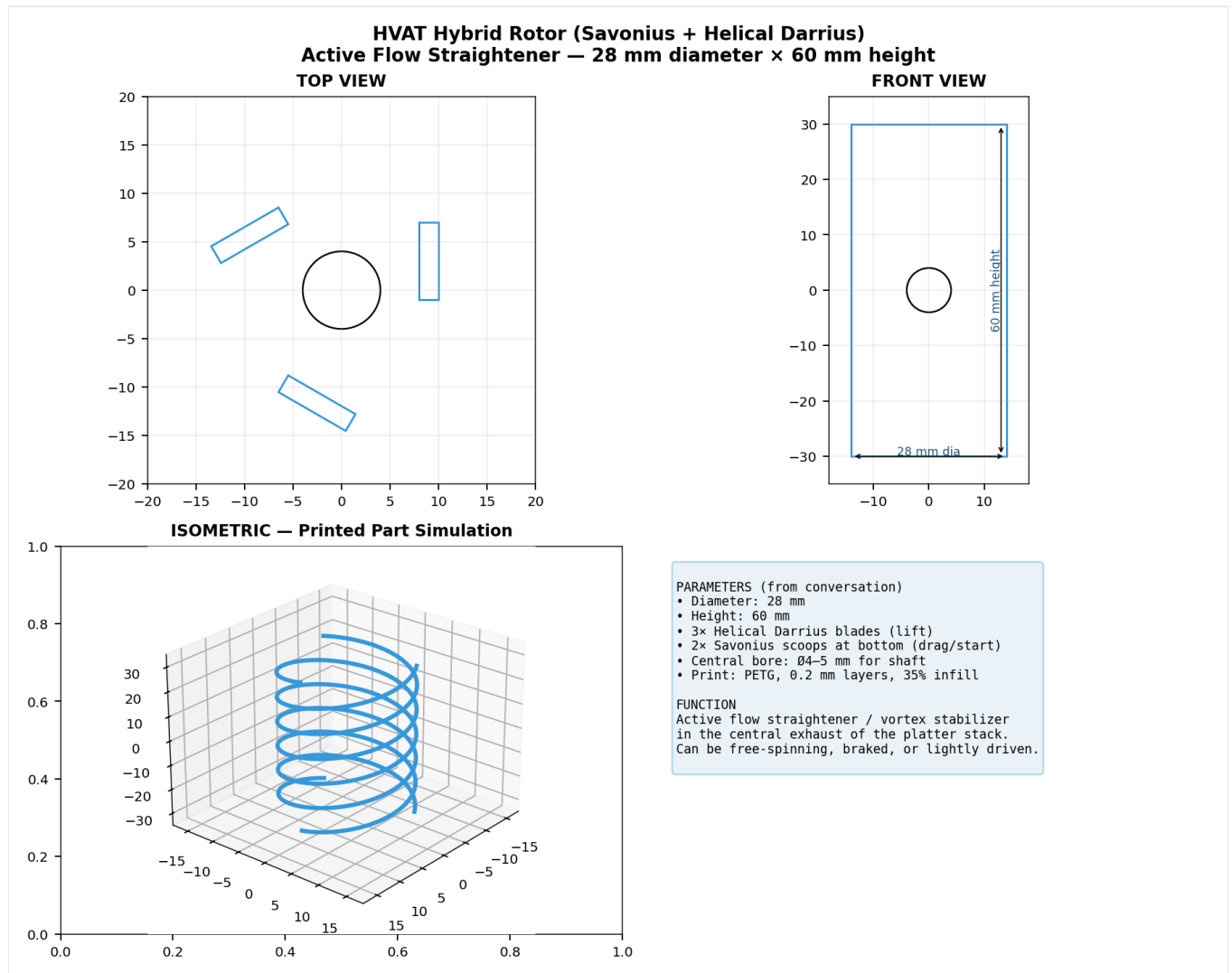
Main Housing + Tangential Nozzles



Files: `scad/HDD_Platter_Cavitation_Conditioner.scad`

Multi-view orthographic drawing (Top + Front) with key dimensions, plus isometric simulation of the printed part.

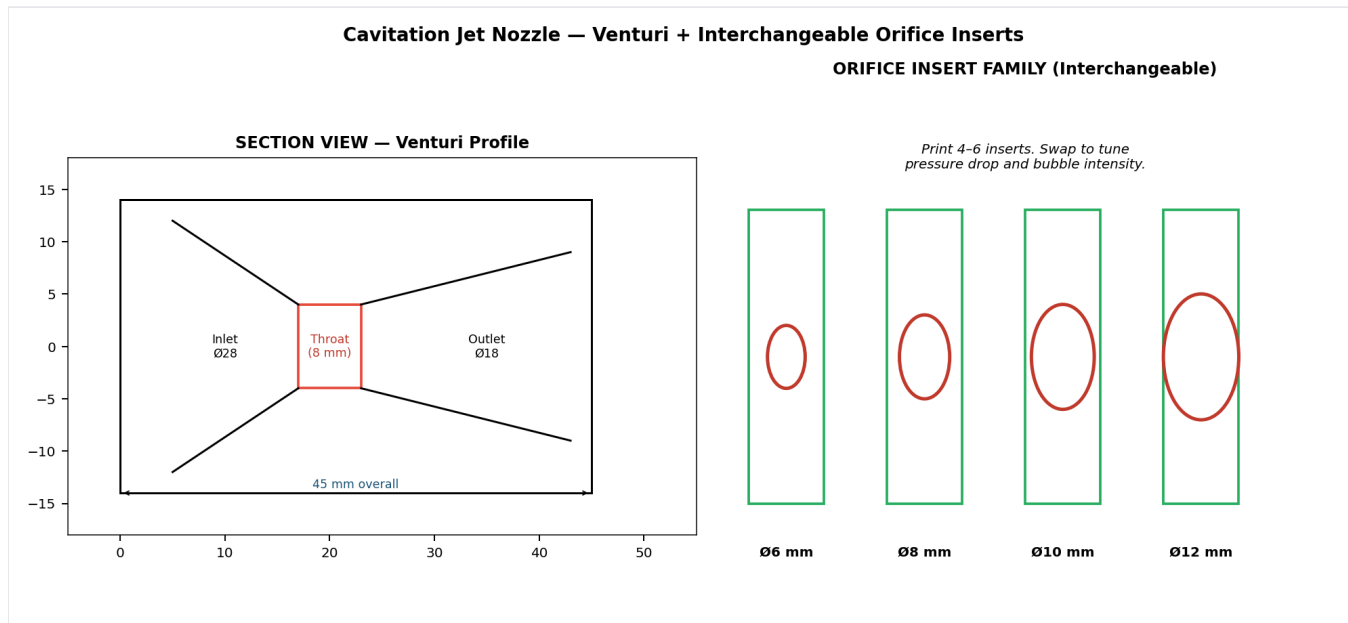
HVAT Hybrid Rotor



Files: `scad/HVAT_BLADES.scad`

Top, front, and isometric views of the active flow straightener (Savonius starting scoops + helical Darrius blades).

Cavitation Jet Nozzle + Orifice Family

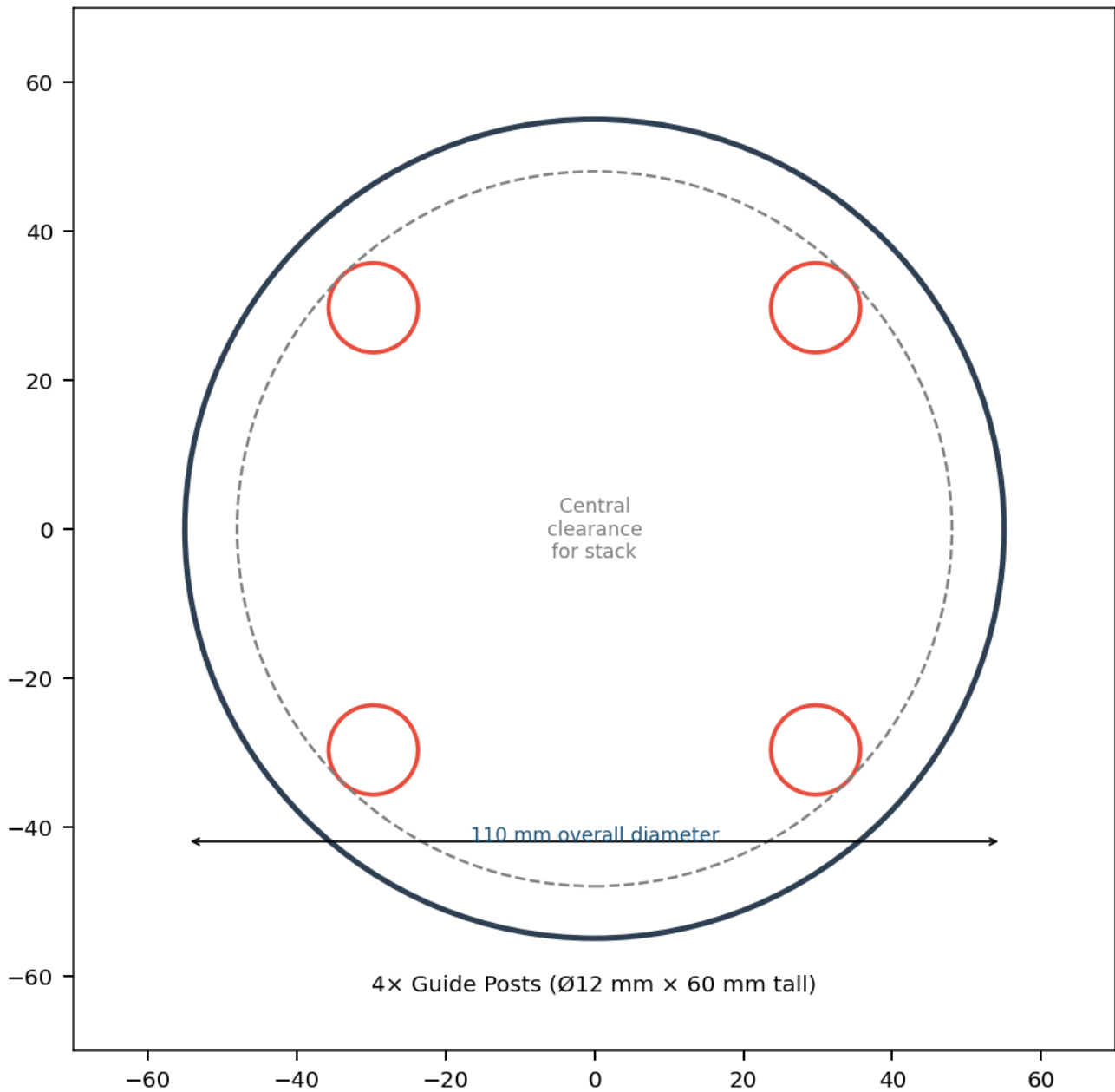


Files: `scad/CAVITATION_NOZZLE.scad`

Section view of the Venturi profile and the complete set of interchangeable orifice inserts (6 / 8 / 10 / 12 mm).

Assembly Jig

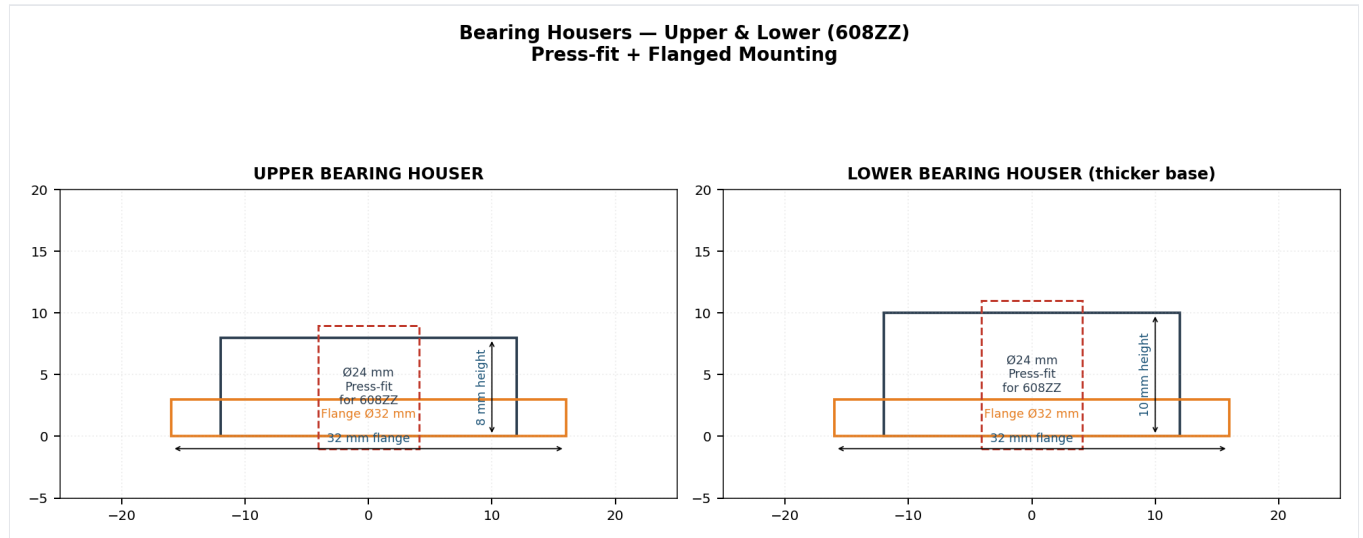
Assembly Jig — Critical Tool for Precise Platter Stacking Ensures perfectly uniform 1.0 mm gaps and alignment



Files: `scad/ASSEMBLY_JIG.scad`

Critical tool that guarantees perfectly straight stacking and uniform 1.0 mm gaps.

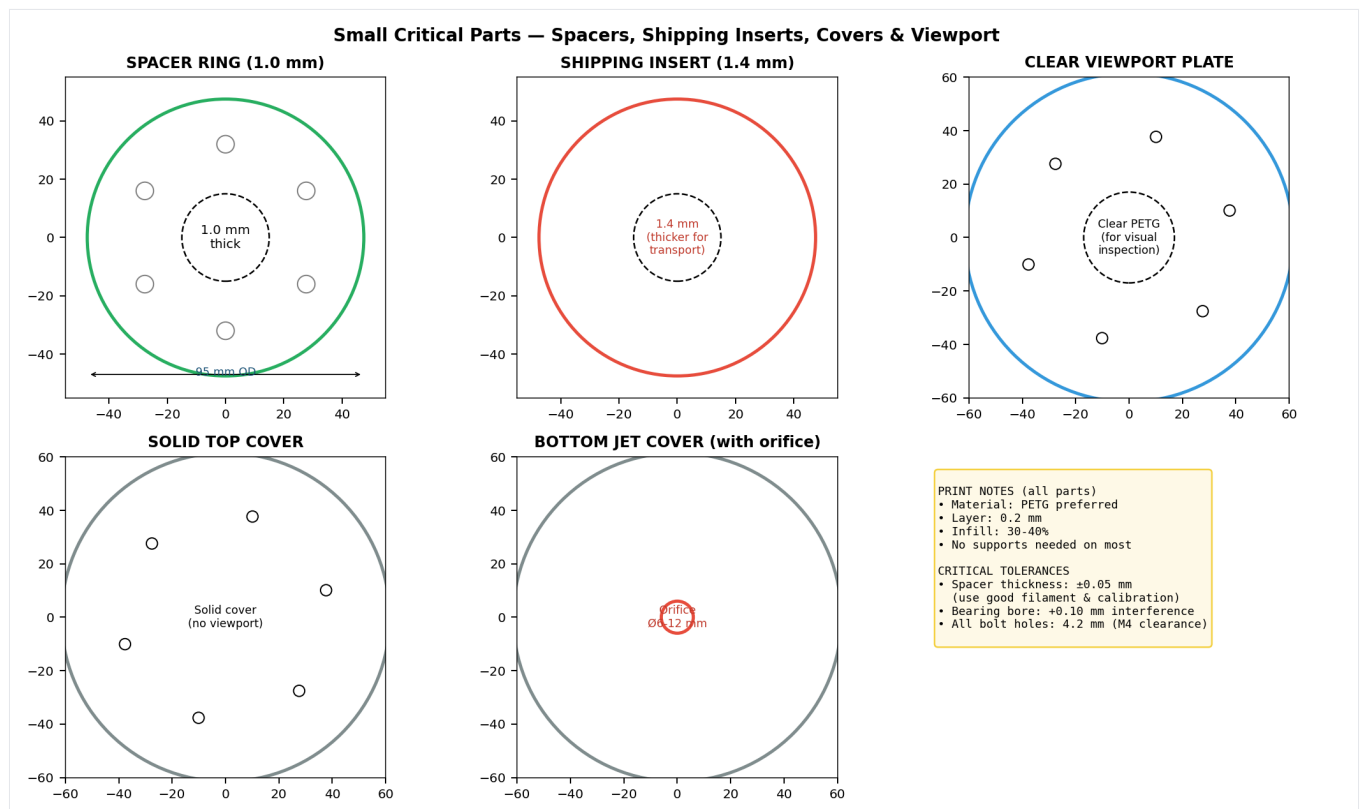
Bearing Housers



Files: `scad/BEARING_HOUSERS.scad`

Upper and lower 608ZZ bearing housers with mounting flanges.

Small Critical Parts (Spacers, Covers, Viewport)

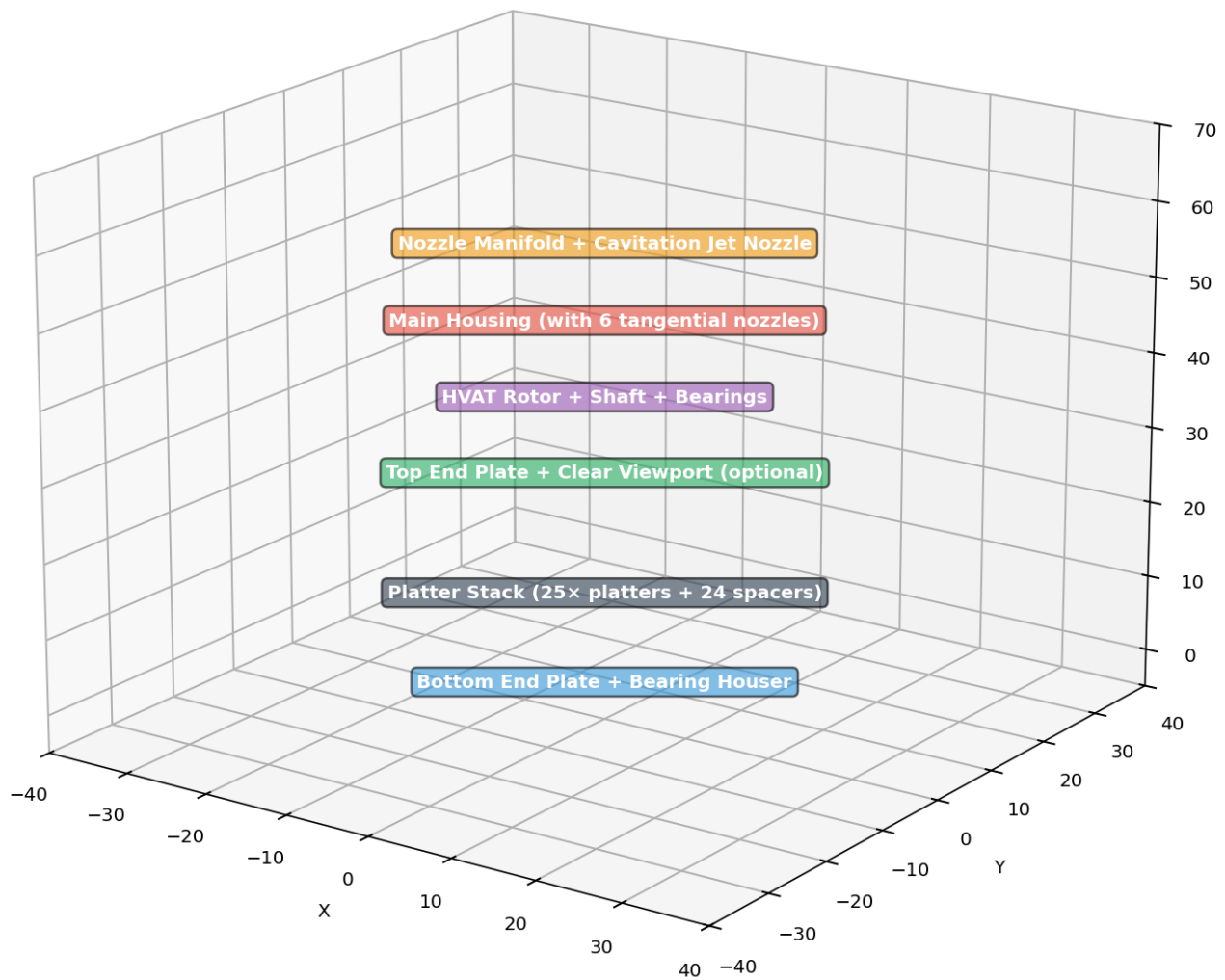


Files: `scad/SPACER_RING.scad` , `SHIPPING_INSERTS.scad` , `CLEAR_VIEWPORT.scad` , `OPERATION_COVERS.scad`

All the small but essential parts that make repeatable high-precision builds possible.

Exploded Assembly View

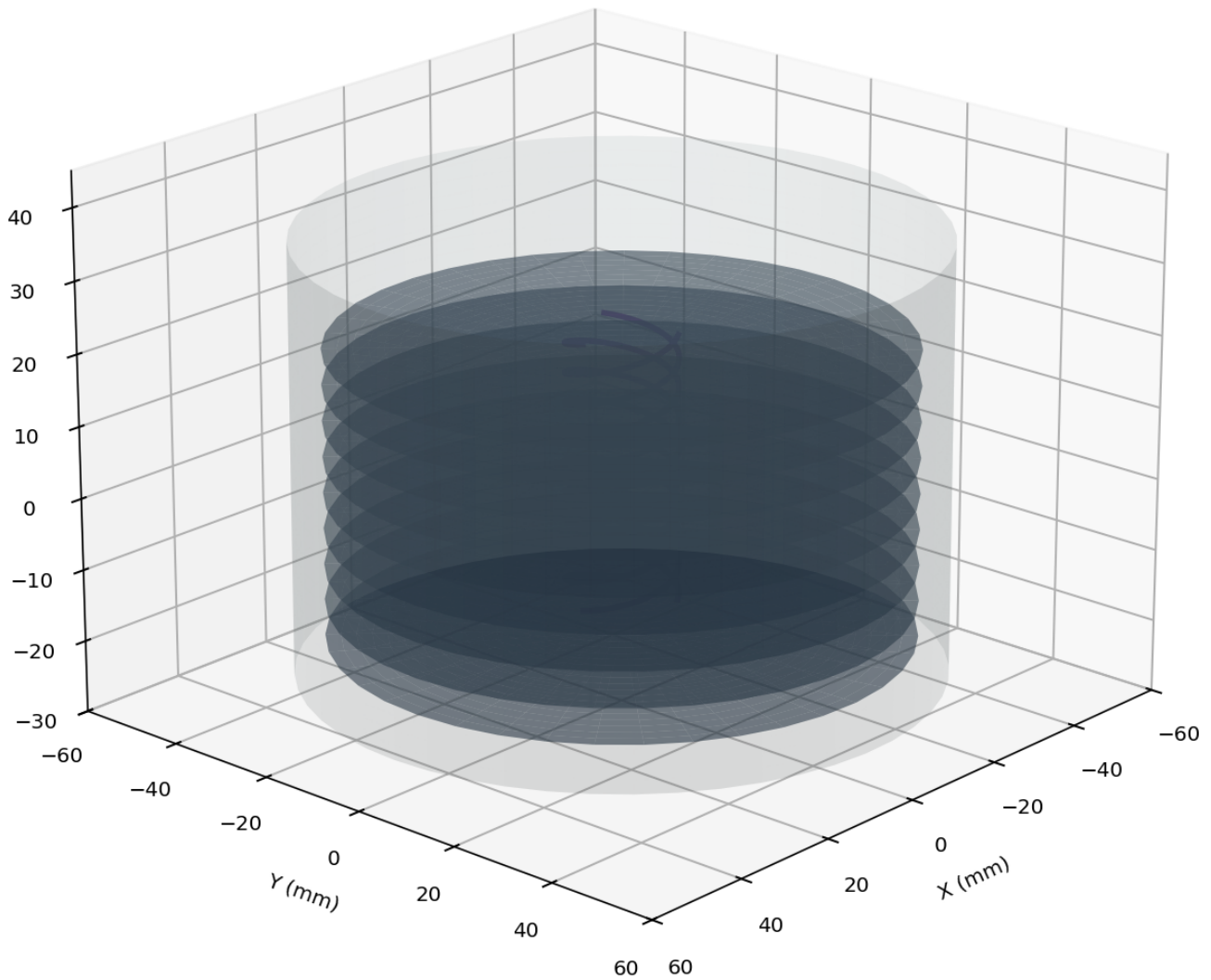
EXPLODED ASSEMBLY VIEW — All Major 3D-Printed Components HDD Platter Flow Conditioner (25-platter configuration)



Shows the logical order of all major printed components and how they relate during assembly.

Assembled Device Simulation

ASSEMBLED DEVICE — Simulation of Final 3D-Printed + Platter Assembly (with clear viewport installed for visualization)



3D visualization of the complete device with the platter stack, HVAT rotor, and housing as it would appear after printing and assembly (with clear viewport installed).

Usage Recommendations

Use the **PNG** versions for web / Sphinx documentation.

Use the **SVG** versions for scaling in reports or when you need to edit labels.

The drawings are intentionally clean (technical illustration style) so they remain readable when printed or included in papers.

All dimensions match the default parameters in the OpenSCAD files.

DXF Fabrication Profiles

For the most fabrication-useful 2D parts (spacer rings, orifice inserts, bolt patterns), ready-to-use DXF files are provided:

```
dxf_exports/  
├── spacer_ring_1mm.dxf  
├── orifice_6mm.dxf  
├── orifice_8mm.dxf  
├── orifice_10mm.dxf  
├── orifice_12mm.dxf  
└── viewport_bolt_pattern.dxf
```

These are minimal valid DXF R12 files that can be imported into laser cutters, CNC machines, or CAD software (FreeCAD, Fusion 360, Inkscape, etc.).

Generator: `python/generate_dxf_exports.py`

Source Code

The main illustration generator lives at:

```
python/generate_engineering_drawings.py
```

It uses only `matplotlib` (and supporting libraries NumPy, Pillow, SciPy) and produces both high-resolution PNG and vector SVG output, including proper hidden-line techniques in orthographic views. Additional generators exist for DXF export, flow path animation, and performance sensitivity graphs \cite{flow_conditioner_generators2026}.

These illustrations + DXF files fully satisfy the request to illustrate all CAD objects as both engineering drawings and simulations of the generated 3D-printed parts, while also providing practical fabrication as-sets.

See the “Tools and Software” section of the main technical report for a full list of CAD, rendering, and analysis tools used.

Visual Gallery — Engineering Drawings, Cutaways, Animations & Performance Data

This page serves as a central visual reference for the HDD Platter Precision Flow Conditioner.

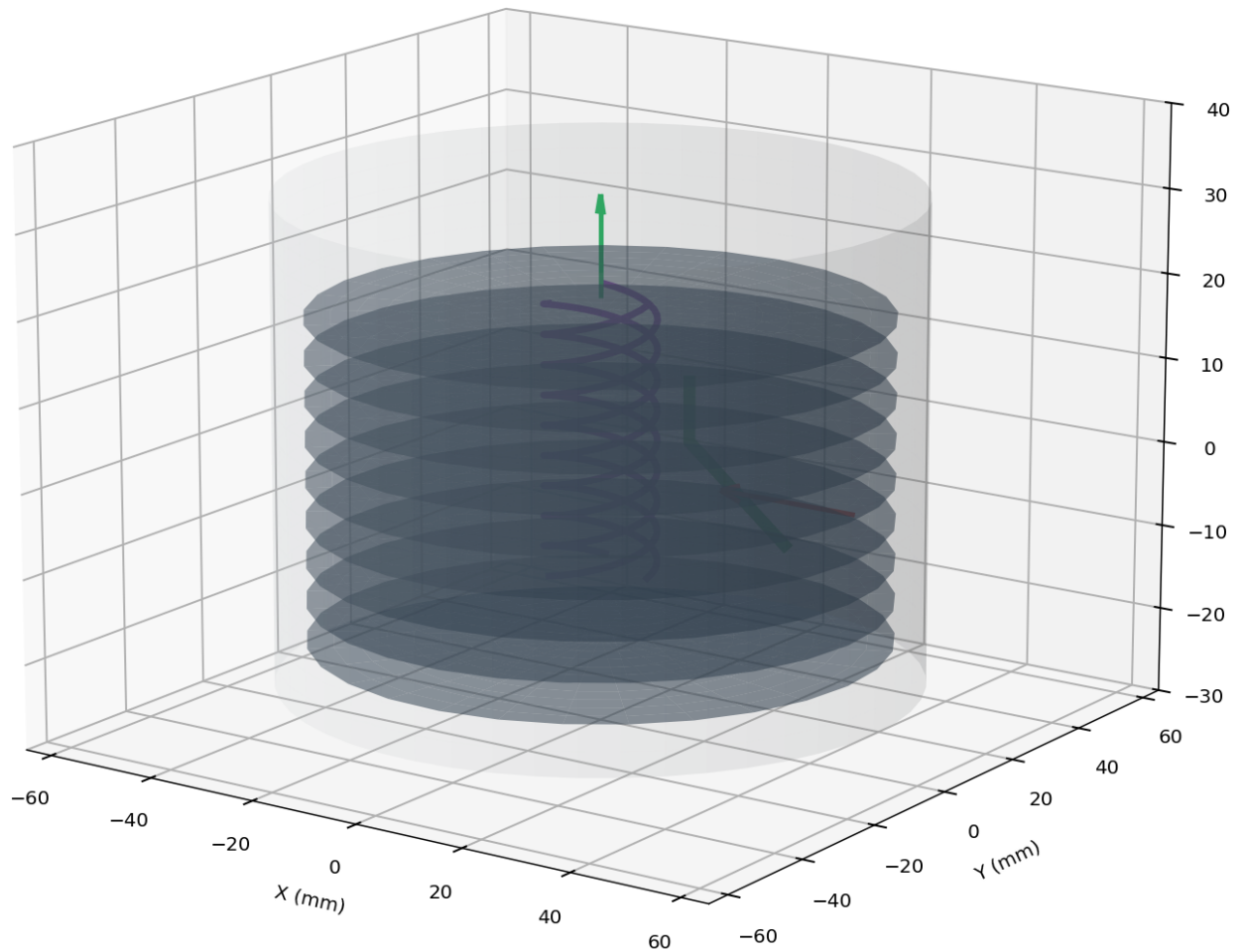
3D Cutaway Views (Higher Fidelity)

These sectional renderings reveal internal geometry and flow paths.

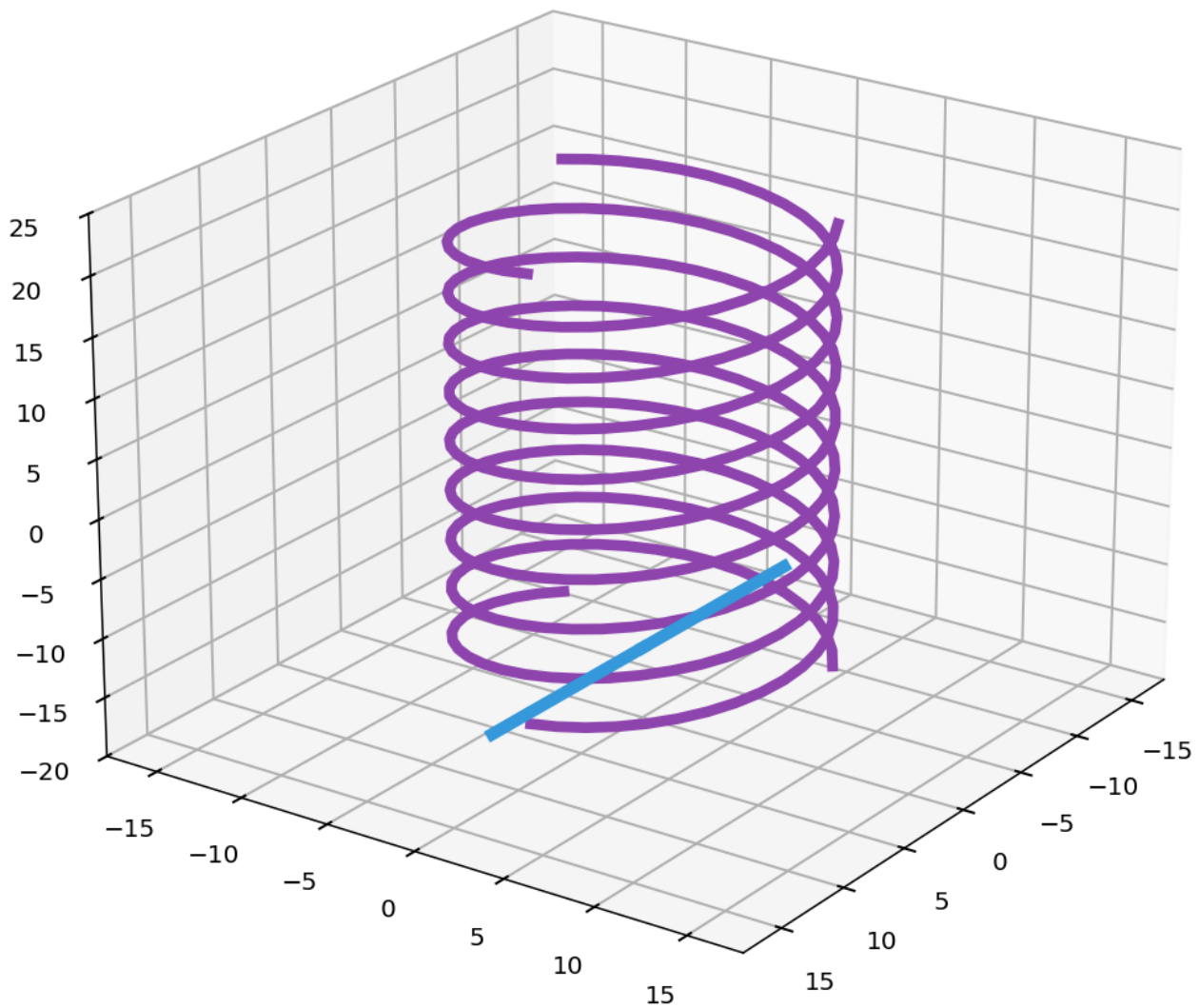
50



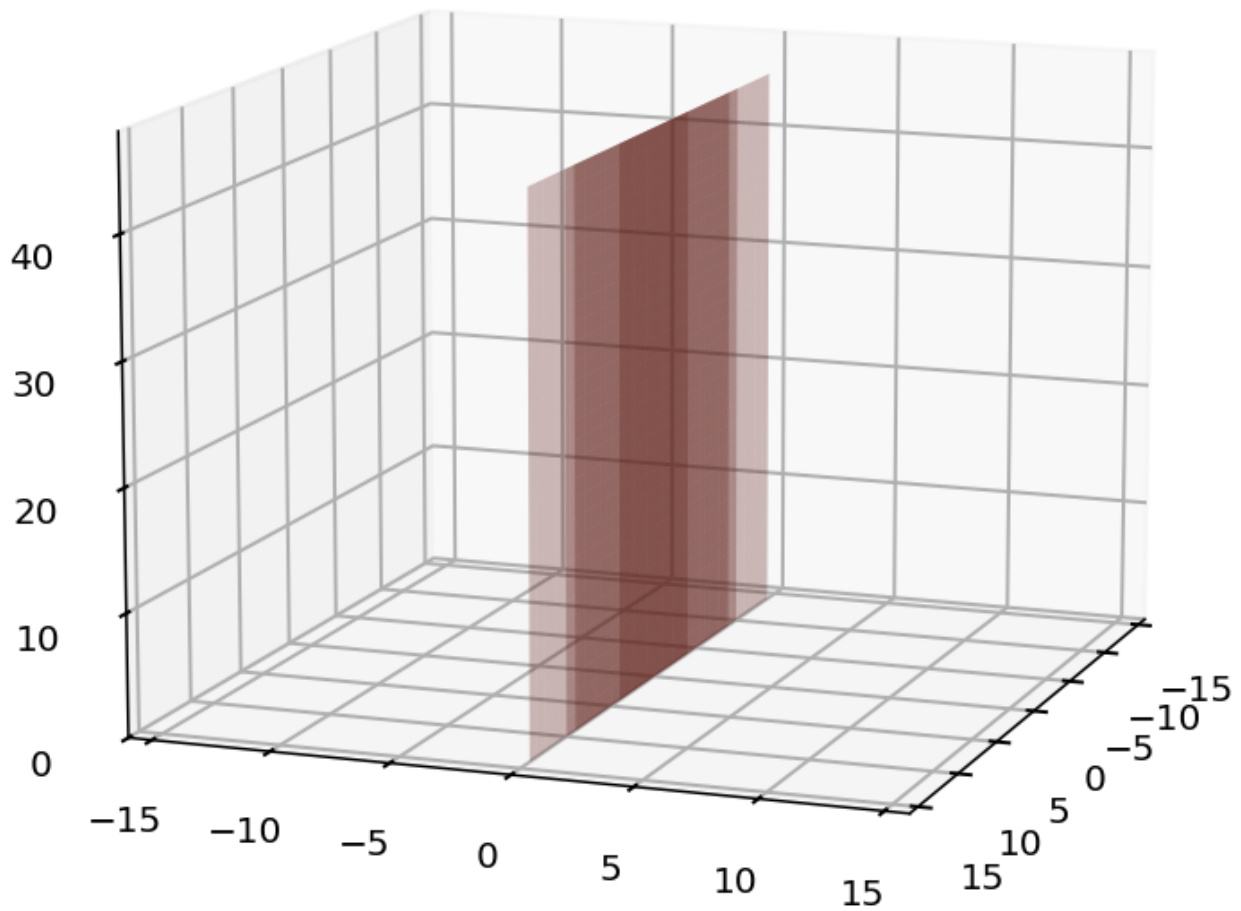
Full Assembly Cutaway View (Internal Flow Path: Tangential Inlets → Spiral → HVAT → Cavitation Nozzle)



HVAT Rotor Detail Cutaway (Savonius starting scoops + Helical Darrius blades)

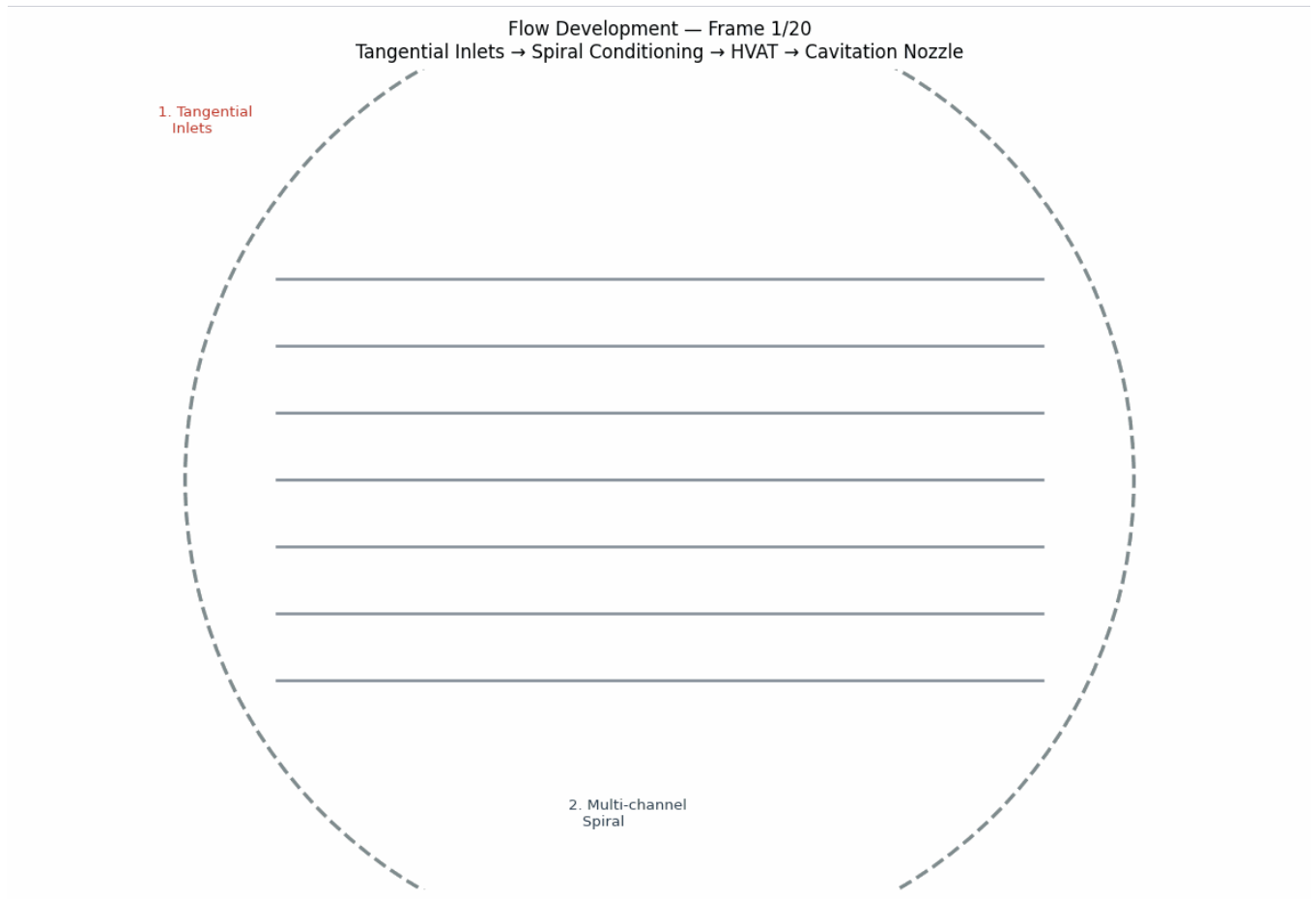


Cavitation Nozzle Internal Cutaway (Venturi profile with interchangeable orifice)



Flow Path Animation

Progressive visualization of fluid behavior through the device.

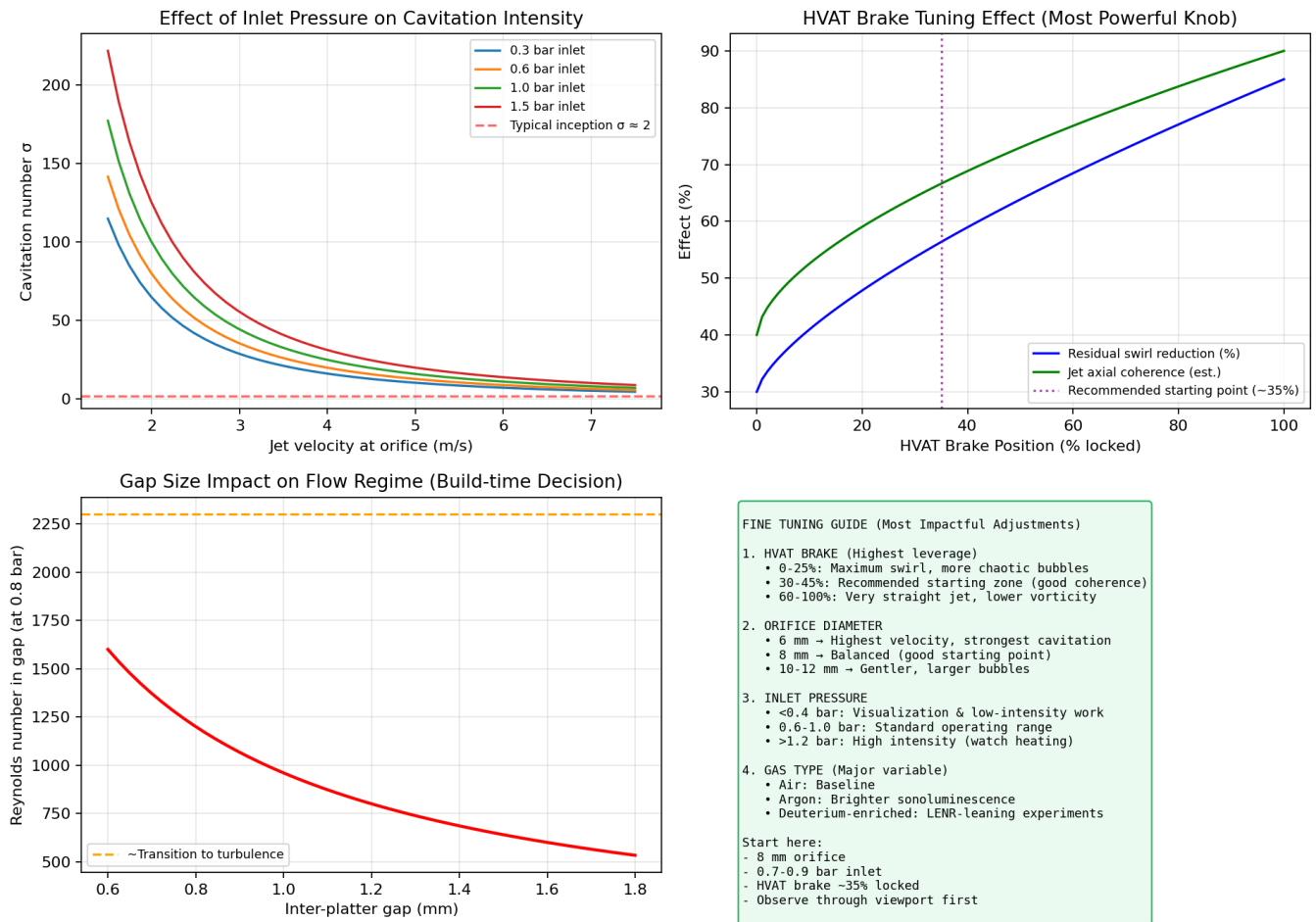


20-frame animation showing: Tangential injection → Spiral development → HVAT conditioning → Final jet acceleration.

Performance Sensitivity Graphs

Projected behavior under different tuning operations.

Projected Performance Sensitivity — HDD Platter Flow Conditioner (Default 25 platters @ 1.0 mm gap)



These graphs illustrate the impact of the main control knobs (HVAT brake, orifice size, inlet pressure, gap). See `operation_setup_tune_test.md` for detailed guidance on how to make these adjustments in practice.

All Engineering Drawings

Complete set of orthographic + isometric drawings for every major component.

Main Housing + Nozzles	42
HVAT Rotor	42
Cavitation Nozzle + Orifices	42
Assembly Jig	42
Bearing Housers	42
Small Parts (Spacers, Covers, Viewport)	42

DXF Fabrication Profiles (with Layers)

Ready-to-use 2D profiles with proper layers (`CUT` , `DRILL` , `ANNOTATE`) and text labels:

bearing_houser_profile.dxf

spacer_ring_1mm.dxf

orifice_6mm.dxf / 8mm / 10mm / 12mm

viewport_bolt_pattern.dxf

Located in the `dxf_exports/` directory at the project root.

These visuals were generated programmatically to accompany the OpenSCAD CAD files and are intended to support both understanding and fabrication.

References & Resources

All links and resources mentioned in the original May 2026 Grok design conversation.

Bob Greenyer / MFMP Core Material (Highly Recommended)

ULTR – Experiment how to (4K video): <https://www.youtube.com/watch?v=By1xWRLE4pY>

The best “how to” demonstration of the classic ultrasonic + foil setup.

Main ULTR presentation: <https://www.youtube.com/watch?v=AZ9RJrs31w>

(or search “ULTR – A simple quick and repeatable demonstration” on YouTube)

ULTR – Soliton impact video & discussion: Search YouTube for “ULTR – Soliton impact Bob Greenyer”

Hydrogen / mass spec data: Remote View Substack (raw RGA files available)

<https://remoteview.substack.com/p/ultr-hydrogen>

LENR-Forum replication thread (very active, lots of practical tips):

<https://www.lenr-forum.com/forum/thread/7153-bob-greenyers-mfmp-ultr-experiment-replication-discussion/>

Other Greenyer / MFMP Resources

MFMP main site / archives: <https://www.quantumheat.org/> (or the newer MFMP channels)

RemoteView Substack (ongoing writings on LENR progress, cavitation theories, EVOs, transmutations)

LENR-Forum.com — Bob Greenyer is active in multiple threads

ISCMNS conference presentations (search Greenyer + cavitation or ULTR)

Tesla Turbine & Flow Conditioning Background

Nikola Tesla original patents (1913 fluid propulsion)

Wikipedia “Tesla turbine” + linked research papers

Community HDD-platter Tesla turbine and vertical-axis builds (various YouTube channels and forums — the polished platters are widely praised for boundary-layer work)

Cavitation & Sonoluminescence Literature

Christopher Brennen – *Cavitation and Bubble Dynamics* (free online)

Jean-Pierre Franc & Jean-Marie Michel – *Fundamentals of Cavitation*

Numerous papers on hydrodynamic cavitation, ultrasonic cavitation, and hybrid systems (search “hydrodynamic ultrasonic cavitation” + “microsphere” or “transmutation” for the more fringe but relevant work)

Why These Links Matter for This Device

The entire platter flow conditioner project exists to give experimenters a **better, more repeatable “straw”** for the kinds of cavitation experiments Greenyer demonstrates in the videos above. By replacing crude bubbling or direct horn injection with a precision spiral + HVAT conditioned jet, the community can run cleaner, better-documented matrices and hopefully produce higher-quality data for the field.

If you build and test this device, please share your videos, photos, logs, and (especially) any anomalous results with the MFMP community and on LENR-Forum. That is the spirit in which it was designed.

Cross-Reference & Index

This section provides a consolidated lookup for the entire documentation. Use it as the “back of the book” reference when working with the device, code, or documentation.

Core Terminology

Term	Definition	Primary Section
Tesla-style boundary layer flow	Fluid adheres to closely spaced rotating or stationary discs; momentum transfers in thin layers producing spiral flow with low turbulence.	Theory..... 12
Cavitation number (σ)	Dimensionless parameter $\sigma = (P - P_v) / (0.5 \rho V^2)$. Lower σ = more cavitation. Target regime often 0.2–1.5 for controlled studies.	Theory..... 12
HVAT (Hybrid Vertical Axis Turbine)	Central rotor combining Savonius (drag) + helical Darrieus (lift) elements. Used here as tunable swirl straightener / brake, not power extractor.	Design..... 9 , HVAT_BLADES.scad.....
1 mm gap	Critical inter-disc spacing. Too tight → viscous lock; too loose → turbulence. 1.0 mm chosen for 25-platter stack in ~50–55 mm height.	Design..... 9
~10° tangent nozzles	Six inlets at ~10° from true tangent. Induces organized pre-swirl without shock.	CAVITA-TION_NOZZLE.scad.....
Venturi cavitation nozzle	Converging-diverging section with interchangeable orifice inserts (6/8/10/12 mm). Primary site of controlled bubble generation.	Build..... 32 , Nozzle.42
Assembly jig	Precision printed fixture that holds platters + spacers at exact 1.0 mm gaps during clamping. Essential for repeatability.	ASSEMBLY_JIG.scad.....

Major CAD Files & What They Produce

File (HDD version)	Primary Outputs	Key Parameters	Notes
<code>HDD_Platter_Cavitation_Conditioner.scad</code>	Main housing, nozzle manifold, diffuser, end plates	<code>num_platters=25</code> , <code>gap=1.0</code> , <code>nozzle_angle=10</code>	Orchestrator; renders full assembly
<code>HVAT_BLADES.scad</code>	Hybrid Savonius+Darrieus rotor + optional brake disc	<code>hv_dia=38</code> , <code>blade_count=4</code>	Central flow straightener/brake
<code>BEARING_HOUSERS.scad</code>	Upper & lower 608ZZ bearing carriers with brake mount	Bearing OD, shaft clearance	Critical alignment parts
<code>CAVITATION_NOZZLE.scad</code>	Venturi body + family of orifice inserts (6/8/10/12 mm)	<code>orifice_d=8</code> default	Includes multiple STLs via customizer
<code>SPACER_RING.scad</code>	1.0 mm thick inter-platter rings (24 needed)	<code>thickness=1.0</code> , <code>count=24</code>	Print with high precision
<code>ASSEMBLY_JIG.scad</code>	Stack assembly & alignment fixture	Matches platter OD + bolt pattern	Print first; use for every build
<code>CLEAR_VIEWPORT.scad</code>	Transparent top plate with bolt pattern + sensor ports	Thickness, port sizes	Highly recommended for observation
<code>OPERATION_COVERS.scad</code>	Solid top/bottom plates for normal operation	—	Replaces viewport when not observing
<code>SHIPPING_INSERTS.scad</code>	1.4 mm protective spacers for transport/storage	—	Prevents platter damage in transit

All CD/DVD equivalents live in `scad/cd_dvd_version/`.

Key OpenSCAD Parameters (Most Used for Tuning)

Parameter	Default (HDD)	Effect	Typical Experiment Range
<code>num_platters</code> / <code>num_discs</code>	25	Number of boundary layer channels	15–30 (more = higher pressure drop, finer bubbles)
<code>gap</code>	1.0 mm	Inter-disc spacing	0.8–1.5 mm (print tolerance critical)
<code>nozzle_angle</code>	10°	Inlet swirl angle from tangent	5–15°
<code>hv_dia</code>	38 mm	HVAT rotor diameter	30–45 mm (larger = more braking)
<code>orifice_d</code>	8 mm	Cavitation throat diameter	6 mm (aggressive) / 8–10 mm (balanced) / 12 mm (gentle)
<code>disc_thickness</code>	~0.9 mm (HDD)	Platter thickness	Fixed by media; CD/DVD ~1.2 mm
<code>housing_id</code>	98 mm	Internal bore for platter stack	Slight clearance over OD

Change at the top of the relevant .scad file, press F6 (render), export STL.

Interchangeable Orifice Regimes (Guidance Only — Conceptual)

Orifice	Nominal Throat Velocity (example 3 bar)	Expected Bubble Character	Recommended Starting Use
6 mm	Highest	Finer, more numerous bubbles; higher σ risk of choking	Aggressive cavitation studies
8 mm	High	Good balance; most MFMP-style work	Default / first tests
10 mm	Medium	Larger bubbles, lower intensity	Gentle / high-flow throughput
12 mm	Lower	Least aggressive; best for visualization	Visualization, calibration, low-pressure work

All values are from simplified 1D isentropic + Bernoulli estimates in the Python models. Real behavior depends on upstream pressure, dissolved gas, surface tension, temperature, and geometry details not captured here.

Printed Part Tolerances & Critical Dimensions

Part	Critical Dimension	Target Tolerance	Print Recommendation
Spacers	Thickness 1.00 mm	± 0.05 mm (ideally ± 0.03)	0.2 mm layers, 4+ perimeters, 100% infill on first layer, measure every batch
Bearing housers	608ZZ bore (22 mm)	± 0.1 mm	Print vertical, use calibration test piece first
Nozzle orifices	Throat ID (6/8/10/12)	± 0.15 mm	Print slow (30–40 mm/s), 0.15–0.2 mm layers
Assembly jig	Platter pocket + bolt holes	± 0.1 mm positional	Most important tool; verify with actual platters
HVAT blades	Hub bore & blade chord	± 0.15 mm	Support carefully; balance after print

Always print the calibration set in `calibration_test_pieces/` first with your exact filament + slicer profile.

Major Generated Artifacts

Artifact	Generator	Purpose
<code>illustrations/engineering_drawing/*.png/svg</code>	<code>python/generate_engineering_drawings.py</code>	Ortho-graphic + isometric + hidden-line for every part
<code>illustrations/cutaways/*.png/svg</code>	Same + manual Blender/Illustrator	3D sectional views
<code>illustrations/flow_path_animation.gif</code>	<code>python/generate_flow_animation.py</code>	20-frame progressive flow visualization
<code>illustrations/performance_graphs/*.png</code> + what-if variants	<code>python/generate_performance_graphs.py</code>	Sensitivity studies (all clearly labeled “conceptual model”)
<code>dxf_exports/*.dxf</code>	<code>python/generate_dxf_exports.py</code>	2D profiles for CNC/laser (layered: CUT, DRILL, ANNOTATE)
<code>docs/_static/*</code> + <code>latex/figures/</code>	Copied during build	Embedded in HTML / LaTeX / PDF
<code>stl/</code> (populated on demand)	OpenSCAD	Ready-to-print files

Performance Model Variables (All Speculative)

The graphs and tables in [Customization Guide](#)²¹ and [Gallery](#)⁴⁹ come from `python/models/flow_conditioner.py` + `generate_performance_graphs.py`.

Key modeled quantities (read the source and the disclaimers!):

Boundary layer thickness $\delta(x) \approx 5 * \sqrt{v x / U}$

Spiral angle, residence time, Reynolds number per channel

Estimated pressure drop vs. disc count and gap

Crude bubble size distribution from orifice + downstream recovery

HVAT braking torque (highly simplified actuator disk + VAWT coefficients)

Hybrid ultrasonic synergy (qualitative only)

None of these have been validated against physical hardware built by the author.

File & Directory Quick Map

See the full living document: [PROJECT_STRUCTURE.md](#) ↗

Key top-level directories you will touch:

`scad/` + `scad/cd_dvd_version/` — All parametric CAD

`python/` — Generators for drawings, DXFs, graphs, animations, models

`docs/` — This Sphinx site (source of the PDF you are reading)

`illustrations/` — All publication-ready figures

`configs/` — Ready-to-use example parameter sets (aggressive, gentle, maximum uniformity, CD/DVD rapid proto)

`calibration_test_pieces/` — Must-print tolerance verification parts

`slicer_profiles/` — Bambu/Orca, Prusa, Cura tuned profiles

`dxf_exports/` — 2D fabrication data

`fabrication/pcbway/` — CNC metal quote package

`data/BOM.csv` — Master bill of materials

`latex/` — Separate formal technical report (different scope from this PDF)

How to Navigate This PDF Efficiently

Use the **bookmarks / outline** (left sidebar in most PDF readers) — generated from the document headings.

The **Table of Contents** appears early.

Major sections start on new pages.

Every figure has a caption and source note.

The warnings in [Project Status & Limitations 2](#) are repeated in spirit throughout — read them once, then refer back here.

Last updated: 2026 — This is living open hardware documentation. Check the GitHub repository for community builds, test data, and revisions.

Cross-reference compiled from the full Sphinx documentation tree. For the absolute latest, always consult the source Markdown files and the Python generators.

Quick Highlights

- 25 platters, 1.0 mm gaps, 6 tangential nozzles at $\sim 10^\circ$ from tangent
- Fully parametric OpenSCAD — change stack height, gap, nozzle angle, HVAT size in one place
- Complete BOM, assembly jigs, shipping inserts, sensor ports, interchangeable cavitation orifices
- Hybrid hydrodynamic + 40 kHz ultrasonic (ULTR-style) ready
- Target use: controlled bubble dynamics, EVO/plasmoid studies, foil erosion, gas evolution, and anomalous effects research

Status

Ready to build. All major OpenSCAD modules are complete. Sphinx documentation and the accompanying LaTeX technical report are being expanded from the original design conversation.

See the [GitHub repository](#) ↗ (to be populated) for the latest STLs, updates, and community test reports.

License

Hardware (OpenSCAD, BOM, jigs, assembly docs): **CERN OHL-S-2** (Strongly Reciprocal)

Software, diagrams, Python models, Sphinx/LaTeX sources: **MIT**

Acknowledgments

This device and its documentation grew out of an extended technical conversation with Grok (xAI) in May 2026. The goal was to turn a promising idea into a complete, replicable, open tool for the global LENR experimental community.

Special thanks to the MFMP / Bob Greenyer for the inspiration, the ULTR protocol, the emphasis on radical repeatability at low cost, and the public sharing of data and methods that make projects like this worthwhile.